Procurement Intelligence & Supplier Optimization Analytics Platform

Integrated Three-Case Study Report

Date: July 2026
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Distribution: CFO, Procurement Director, AP Leadership, Category Managers

Platform Overview

Business Problem

Procurement teams manage hundreds of suppliers and thousands of invoices but lack visibility into supplier risk, purchasing opportunities, and vendor performance. Most organizations track suppliers in silos—the Vendor Master holds risk ratings, the Accounts Payable (AP) system holds invoice errors, and the Purchasing system holds order history. Rarely are these datasets merged to answer critical questions about supplier health, contract suitability, and portfolio optimization.

Objective

Build an analytics solution that helps procurement leaders answer:

Question	Project
Which suppliers are risky?	Project 1: Supplier Risk Assessment
Which purchases should move to strategic contracts?	Project 2: Strategic Contract Optimization
Which suppliers deserve preferred status?	Project 3: Supplier Portfolio Optimization

The Outcome of Each of Those Studies Will Provide:

This report is strategically structured to guide stakeholders through a cohesive and actionable narrative. Unlike traditional dashboards, where charts often stand alone as isolated visualizations, this report ensures that each chart complements the others, creating a seamless flow of insights.

Here's what you'll find:

- A Purposeful Progression: Each visualization is designed to answer a specific business question, building on the insights of the previous one. This ensures that the report doesn't just present data—it tells a story that aligns with executive priorities.
- Decision-Driven Insights: The charts are organized to naturally feed into one another, guiding you from identifying key suppliers to understanding financial exposure, assessing risks, and ultimately determining actionable next steps.
- Consulting-Grade Clarity: Following the best practices of top consulting firms, every chart includes a clear purpose, rationale, insights, and recommended actions, making it easy to extract value and make informed decisions.

By the end of this report, you'll have a holistic view of the supplier landscape, backed by data-driven recommendations that are ready for execution.

Engine	**Question**	**Output**
Supplier Risk Management	Which suppliers create operational risk?	Retain / Monitor / Replace

Contract Intelligence	Which products should be under contract?	Contract / Review / Phase-Out
Supplier Development	Which suppliers deserve more business?	Strategic / Preferred / Approved / Monitor

Platform Architecture

Data Flow Diagram

```text
┌──────────────────────────────────────────────────────────────┐
│                         ERP SYSTEM                           │
│  ┌────────────┐  ┌────────────┐  ┌───────────────────────┐  │
│  │ Vendor      │  │ Invoice     │  │ Purchase Order         │ │
│  │ Master      │  │ Headers     │  │ History                │ │
│  └────────────┘  └────────────┘  └───────────────────────┘  │
│         │                 │                   │               │
│         └────────────────┼──────────────────┘               │
│                           ▼                                  │
│                 ┌────────────────────┐                       │
│                 │   SQL Views / CTEs  │                      │
│                 │  (Clean, Aggregated)│                      │
│                 └────────────────────┘                       │
└──────────────────────────────────────────────────────────────┘
                           │
                           ▼
┌──────────────────────────────────────────────────────────────┐
│                    ANALYTICS LAYER                            │
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│                                                                  │
│  ┌─────────────────┐  ┌─────────────────┐  ┌─────────────────┐  │
│  │ Risk Management │  │ Contract        │  │ Supplier        │  │
│  │ Engine          │  │ Intelligence    │  │ Development     │  │
│  │                 │  │ Engine          │  │ Engine          │  │
│  │ • Spend         │  │ • Spend Impact  │  │ • Importance    │  │
│  │ • Error Rate    │  │ • Frequency     │  │ • Quality       │  │
│  │ • Rejection     │  │ • Consistency   │  │ • Efficiency    │  │
│  │ • Risk Level    │  │ • Volatility    │  │ • Position      │  │
│  │                 │  │ • Concentration │  │ • Risk          │  │
│  └─────────────────┘  └─────────────────┘  └─────────────────┘  │
│           │                    │                    │           │
│           ▼                    ▼                    ▼           │
│  ┌─────────────────┐  ┌─────────────────┐  ┌─────────────────┐  │
│  │ Retain/Monitor/ │  │ Contract/Review/│  │ Strategic/      │  │
│  │ Replace         │  │ Phase-Out       │  │ Preferred/      │  │
│  │                 │  │                 │  │ Approved/Monitor│  │
│  └─────────────────┘  └─────────────────┘  └─────────────────┘  │
└──────────────────────────────────────────────────────────────────┘
                                   │
                                   ▼
┌──────────────────────────────────────────────────────────────────┐
│                       EXECUTIVE DASHBOARD                        │
│                                                                  │
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│  ┌────────────────────────────────────────────────────────────┐│
│  │  KPI Cards: 88 Suppliers | $627M Spend | $2.7M Savings     ││
│  └────────────────────────────────────────────────────────────┘│
│  ┌────────────────────────────────────────────────────────────┐│
│  │  Supplier Risk Matrix  │  Contract Priority Scatter        ││
│  └────────────────────────────────────────────────────────────┘│
│  ┌────────────────────────────────────────────────────────────┐│
│  │  Supplier Portfolio Quadrant  │  Concentration Heatmap     ││
│  └────────────────────────────────────────────────────────────┘│
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┌────────────────────────────────────────────────────────────────┐
│                      PROCUREMENT DECISIONS                       │
│                                                                  │
│  ✅ Expand relationship with Vintage Store                       │
│  ✅ Negotiate Power Tools contract                               │
│  ✅ Develop backup suppliers for 10 concentration-risk vendors   │
│  ⚠️ Executive review for Accenture                               │
│  ⚠️ Performance Improvement Plan for Fastenal                    │
└────────────────────────────────────────────────────────────────┘
```

Data Model

Simplified Entity-Relationship Diagram

```text
┌─────────────────┐          ┌─────────────────┐
│    VENDORS      │          │    INVOICES     │
├─────────────────┤          ├─────────────────┤
│ vendor_id (PK)  │◄─────────│ vendor_id (FK)  │
│ vendor_name     │          │ invoice_id (PK) │
│ currency        │          │ total           │
│ risk_level      │          │ invoice_date    │
└─────────────────┘          │ validation_status│
         │                   │ payment_status  │
         │                   │ approved_at     │
         │                   └─────────────────┘
         │                            │
         │                            │
         │                   ┌─────────────────┐
         │                   │  INVOICE LINES  │
         │                   ├─────────────────┤
         │                   │ invoice_id (FK) │
         │                   │ line_id (PK)    │
         │                   │ description     │
         │                   │ unit_price      │
         │                   │ quantity        │
```

```
│                    │ line_total     │
└──────────────────▶ └────────────────┘
                              │
                              │
                     ┌────────────────┐
                     │    PRODUCTS     │
                     ├────────────────┤
                     │ product_id (PK) │
                     │ product_name    │
                     │ category        │
                     └────────────────┘
```

Key Relationships:

- One vendor → Many invoices
- One invoice → Many invoice lines
- One invoice line → One product
- One product → Many invoice lines

KPI Definitions

Master KPI Reference

KPI	Definition	Owner	Target

Total Spend	Sum of all invoice totals (USD)	Procureme nt	>$600M
Invoice Accuracy	(Total Invoices - Rejected) / Total Invoices × 100	Supplier	>95%
Payment Rejection Rate	Rejected Payments / Total Invoices × 100	AP	<10%
Vendor Risk Score	Weighted composite (Spend 30% + Errors 30% + Rejections 20% + Risk 20%)	Procureme nt	<40 = Retain
Contract Priority Score	Weighted composite (Spend 30% + Frequency 25% + Consistency 20% + Volatility 15% + Concentration 10%)	Procureme nt	>70 = Contract

Supplier Performance Score	Weighted composite (Importance 20% + Quality 30% + Efficiency 20% + Position 20% + Risk 10%)	Procurement	>70 = Preferred
Price Volatility (CoV)	STDDEV(unit_price) / AVG(unit_price) × 100	Procurement	<20% = Stable
YoY Growth	(Spend_2025 - Spend_2024) / Spend_2024 × 100	Procurement	>10% = Growing

Case Study 1: Supplier Risk Assessment

Project Phase: 1 – Data Exploration & Risk Scorecard Development
Date: July 2026
Author: Ebhota Oje

Distribution: Procurement Director, Accounts Payable (AP) Leadership

1. Executive Summary

This section provides a high-level overview of the business problem, analytical approach, key findings, and the most critical insight for procurement and AP leadership. It is designed to be read in under two minutes.

Block	Summary
Problem	88 vendors, $627M spend. Static risk labels don't capture operational friction.
Approach	Weighted scorecard: Spend 30% + Errors 30% + Rejections 20% + Risk 20%.

Outcome	18 vendors (28% of spend) flagged for monitoring. Zero vendors >70.
Recommendation	Audit Accenture (10.7% errors). Investigate CBRE (56% rejections).

🟢 Hero Finding

Legacy risk labels incorrectly identified the safest supplier as "High Risk."

Observation: Vintage Store is the sole "High Risk" vendor in the master data, yet it has a 0% error rate and 3.6% rejection rate—both significantly below the averages.

Evidence: Transactional performance shows Vintage Store scores 38.5 (Retain), placing it in the safest category.

Investigation: Review the legacy "High Risk" classification. Terminating this vendor based solely on its risk label would be a strategic error.

The Business Problem

Procurement manages 88 active vendors transacting in US dollars, processing 2,168 invoices annually and representing $626.9M in spend. Current supplier oversight relies on static, manually-assigned risk labels ("Low," "Medium," "High") stored in the Enterprise Resource Planning (ERP) vendor master. These labels do not capture operational friction—invoice validation errors and payment rejections—which drive Accounts Payable (AP) inefficiency, delay month-end closes, and strain supplier relationships.

The Analytical Approach

We developed a weighted vendor risk scorecard using transactional data from the ERP. The score combines four components:

Component	Weight	Rationale
Spend Exposure	30%	Higher spend = greater financial materiality
Invoice Error Rate	30%	Validation failures increase AP manual effort
Payment Rejection Rate	20%	Payment friction delays supplier payments
Inherent Risk Level	20%	Legacy master-data classification

Scores are calculated out of 100. Decision thresholds are set at:

- ≥70 → Replace (phase out over 6 months)
- 40–69 → Monitor (Quarterly Business Reviews / improvement plans)
- <40 → Retain (standard oversight)

The Outcome

- 18 vendors (20% of the supplier base) fall into the Monitor zone, representing $174.8M—28% of total spend. These are the highest-priority intervention targets.
- Zero vendors scored 70 or above, indicating the portfolio is operationally stable but requires active performance management.
- The sole "High Risk" vendor (Vintage Store, $6.5M) scored 38.5 (Retain) with a 0% error rate and 3.6% rejection rate, demonstrating that legacy risk labels alone are insufficient for decision-making.
- Conversely, Accenture PLC—classified as "Low Risk"—drives the highest invoice error rate in the dataset at 10.71% (4.7× above the vendor average of 2.29%) , making it the single largest operational friction point.
- CBRE Group (a commercial real estate services firm) has a 56.25% payment rejection rate—the highest in the dataset—signaling an urgent internal process review (remittance, approval workflows, or Purchase Order [PO] matching).

Core Insight: *"Low Risk" does not mean "Low Friction." A dynamic, transaction-based scorecard is essential for effective procurement oversight and AP process improvement.*

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2. Methodology

This section explains the analytical framework in detail. Each design decision is accompanied by its business rationale so that the methodology is defensible and transparent.

2.1 Data Sources & Scoping

To ensure consistency with US-dollar financial reporting and to remove foreign exchange (FX) volatility, we filtered to vendors whose primary currency is USD. This yielded the following scope:

Source	Table	Description

Vendor Master	`vendors_july_2026`	88 active vendors with USD primary currency
Invoice Transactions	`invoice_july_2026`	2,168 invoices linked to these vendors
Scope Filter	`v.currency = 'USD'`	Isolates US-dollar reporting base
Total Analyzed Spend		$626,938,251

2.2 Metrics Calculated

Metric	Definition	Business Relevance
Total Spend	SUM(invoice.total)	Financial exposure and materiality
Invoice Count	COUNT(invoice_id)	Transactional volume / AP workload indicator

Error Rate (%)	(Validation Status = 'Rejected' / Total Invoices) × 100	Invoice quality and compliance from the supplier
Rejection Rate (%)	(Payment Status = 'Rejected' / Total Invoices) × 100	Payment processing friction (remittance, approvals, PO matching)

2.3 Weighted Scoring Model

Component	**Weight**	**Scoring Logic**
Spend Exposure	30%	Normalized against the highest-spend vendor (Michelin = 30 pts).
Invoice Error Rate	30%	Capped at 30%; Score = (Error% / 30) × 30.
Payment Rejection Rate	20%	Capped at 30%; Score = (Rejection% / 30) × 20.

Inherent Risk Level	20%	High = 20 pts, Medium = 10 pts, Low = 0 pts.

2.4 Decision Thresholds

Score Range	Classification	Required Action
≥70	🔴 Replace	Initiate phase-out and competitive Request for Proposal (RFP) process
40–69	🟡 Monitor	Quarterly Business Reviews (QBRs) with structured improvement targets
<40	🟢 Retain	Standard annual performance review

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3. Procurement Spend & Risk Overview

Before assessing individual vendor performance, we aggregated the complete dataset to establish the total addressable spend and overall volume. This provides the macro-level context against which individual vendor scores are compared.

3.1 Overall Procurement Statistics

The table below summarizes the full scope of the analysis. Of the 88 active vendors, 18 are flagged for active monitoring—representing 28% of all spend.

Metric	Value
Active USD Vendors	88
Total Invoices Analyzed	2,168
Total Annual USD Spend	$626,938,251
Vendors in Monitor (Score 40–69)	18 vendors (20%) – $174.8M (28% of spend)
Vendors in Retain (Score <40)	70 vendors (80%) – $452.1M (72% of spend)
Vendors in Replace (Score ≥70)	0

3.2 Distribution of Legacy Risk Labels

The vendor master currently classifies suppliers into three risk tiers. The table below shows how these labels are distributed across the vendor base and, more importantly, how much spend is associated with each label. Notably, "Low Risk" vendors account for over three-quarters of total expenditure, making operational performance within this group particularly consequential.

Risk Level	Vendors	Total Spend	% of Total Spend
Low	68	$478,610,330	76.3%
Medium	19	$141,751,368	22.6%
High	1	$6,576,553	1.0%

3.3 Operational Benchmarks

To contextualize individual vendor performance, we calculated the average error and rejection rates across all 88 vendors. These figures serve as operational baselines; vendors exceeding these averages are prioritized for further investigation.

Metric	Benchmark (Average Across All Vendors)
Invoice Error Rate	2.29%
Payment Rejection Rate	18.76%

4. Key Findings

This section presents the results of the analysis. Each finding is presented with supporting data and interpreted through the lens of the scoring model.

4.1 Priority Vendors – The Monitor Cohort

The table below lists the top 10 vendors from the Monitor cohort (scoring 40–69), ordered by their composite risk score. This represents the highest-priority intervention group, combining financial materiality with operational friction.

Rank	Vendor	Spend	Legacy Risk	Error %	Rejection %	Score	Primary Driver
1	Michelin North America	$12.2M	Medium	3.13%	25.00%	59.8	Highest spend + 1-in-4 invoices rejected
2	Ingersoll Rand	$10.2M	Medium	2.94%	29.41%	57.7	Rejection rate 1.6× above average

3	Alpha Industries	$10.6M	Medium	3.85%	19.23%	52.7	Moderate friction across all metrics
4	Elite Services	$11.5M	Low	5.71%	25.71%	51.1	Low risk label but 2.5× average error rate
5	Caterpillar Inc.	$9.7M	Medium	0.00%	24.14%	50.0	Zero errors, but payment rejections elevated
6	Snap-on Incorporated	$12.0M	Low	3.13%	25.00%	49.3	2nd highest spend, elevated rejections

7	Maersk Line	$10.6M	Medium	0.00%	19.35%	49.0	Moderate rejection, no errors
8	Regal Beloit	$8.9M	Medium	0.00%	24.24%	48.1	Elevated rejection, zero errors
9	C.H. Robinson	$11.6M	Medium	0.00%	14.29%	48.0	Below-average rejection—closer to "Retain"
10	Fastenal Company	$7.9M	Low	7.41%	33.33%	46.9	Highest combined friction in Low-risk group

4.2 Diagnostic Insights – Patterns and Hypotheses

A. Accenture PLC – A "Low Risk" Vendor with "High Risk" Operational Performance

- Context: Accenture is classified as "Low Risk" in the master data. However, its invoice error rate of 10.71% is 4.7× above the 2.29% vendor average. This is the highest error rate in the entire dataset.
- Hypothesis: This pattern is consistent with systemic issues in Accenture's invoice template, PO matching logic, or tax ID validation. An audit of their recent rejected invoices is warranted to isolate the root cause.

B. CBRE Group – The Rejection Outlier

- Context: CBRE Group has a 56.25% payment rejection rate (9 out of 16 invoices rejected). This is 3× above the 18.76% average.
- Hypothesis: This extreme rejection rate is consistent with a single systemic failure—such as outdated remittance instructions, incorrect PO references on the invoice, or an approval workflow bottleneck on our side. AP should prioritize a review of CBRE's payment routing and internal approval chains.

C. Fastenal Company – The Double Threat

- Context: Fastenal exhibits both elevated errors (7.41%, 3.2× average) and elevated rejections (33.33%, 1.8× average), making it the highest combined friction vendor among "Low" risk suppliers.
- Hypothesis: This suggests both invoice quality issues on Fastenal's side and payment processing issues on our side. A comprehensive performance improvement plan addressing both dimensions is recommended.

D. Vintage Store – The "High Risk" Misclassification

- Context: Vintage Store is the sole "High Risk" vendor, yet it has a 0% error rate and 3.57% rejection rate—both significantly below the averages. It scored 38.5 (Retain).
- Hypothesis: The "High Risk" label is not supported by transactional performance. Terminating this vendor based solely on its legacy risk label would be a strategic error and suggests the master-data risk classification should be reviewed for this supplier.

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5. Visualizations Blueprint (Phase 2 – Dashboard Design)

This section provides a blueprint for the Tableau dashboard that will visualize the supplier risk assessment model. Each chart is designed to answer a specific executive question.

Chart 1: Supplier Risk Matrix

Introduction

Purpose of This Chart

The Supplier Risk Matrix is the centerpiece of the risk assessment dashboard. While the treemap (Chart 2) answers *"How much of our money is exposed?"* this chart answers the more immediate operational question: *"Which suppliers deserve our attention right now?"*

A procurement executive needs to prioritize interventions across a portfolio of 88 vendors. The scatter plot is the ideal visualization for this because it plots suppliers along two critical dimensions simultaneously:

- X-Axis (Spend) represents financial exposure—how much we stand to lose if this supplier fails
- Y-Axis (Risk Score) represents operational friction—how much manual effort this supplier requires

This dual-axis encoding immediately reveals which suppliers sit in the Priority Intervention Zone (High Spend + High Risk)—the suppliers that demand immediate executive attention.

Why We Chose a Scatter Plot

Procurement isn’t just about numbers—it’s about making smart trade-offs. Every decision balances financial exposure (how much we spend) with operational risk (what could go wrong). That’s why we turned to a **scatter plot**: it’s the only visualization that lets us see both dimensions at once, making the trade-offs impossible to ignore.

But we didn’t stop there. We added **reference lines** at 40 (Monitor) and 70 (Replace) to create clear decision thresholds. Unlike other charts, this setup doesn’t just show data—it tells you *exactly* what to do next. No ambiguity, no guesswork.

And because procurement is rarely two-dimensional, we introduced **bubble sizes** to represent error rates. Now, at a glance, you can spot which high-risk suppliers are also the most operationally problematic—the ones keeping your team up at night.

Color takes it further. **Green, Amber, Red**—simple, intuitive, and actionable. Retain. Monitor. Replace. These are the decisions procurement leaders face daily, and the scatter plot puts them front and center.

We also kept the focus sharp. Only the most critical suppliers are labeled, mirroring how executives actually review data: they scan for exceptions, the outliers that demand attention.

Finally, we anchored everything with an **Average Spend line**, giving immediate context. Is a supplier above or below the norm? The answer is instant, and the implications are clear.

In short, the scatter plot doesn’t just present data—it drives decisions.

What This Chart Reveals

At its core, this scatter plot answers the critical question every executive asks: *"Which suppliers are putting us at the greatest financial and operational risk?"*

Here’s how it tells the story:

The **top-right quadrant** is where the action is. These are your **priority suppliers**—the ones with high spend *and* high risk. They demand immediate attention.

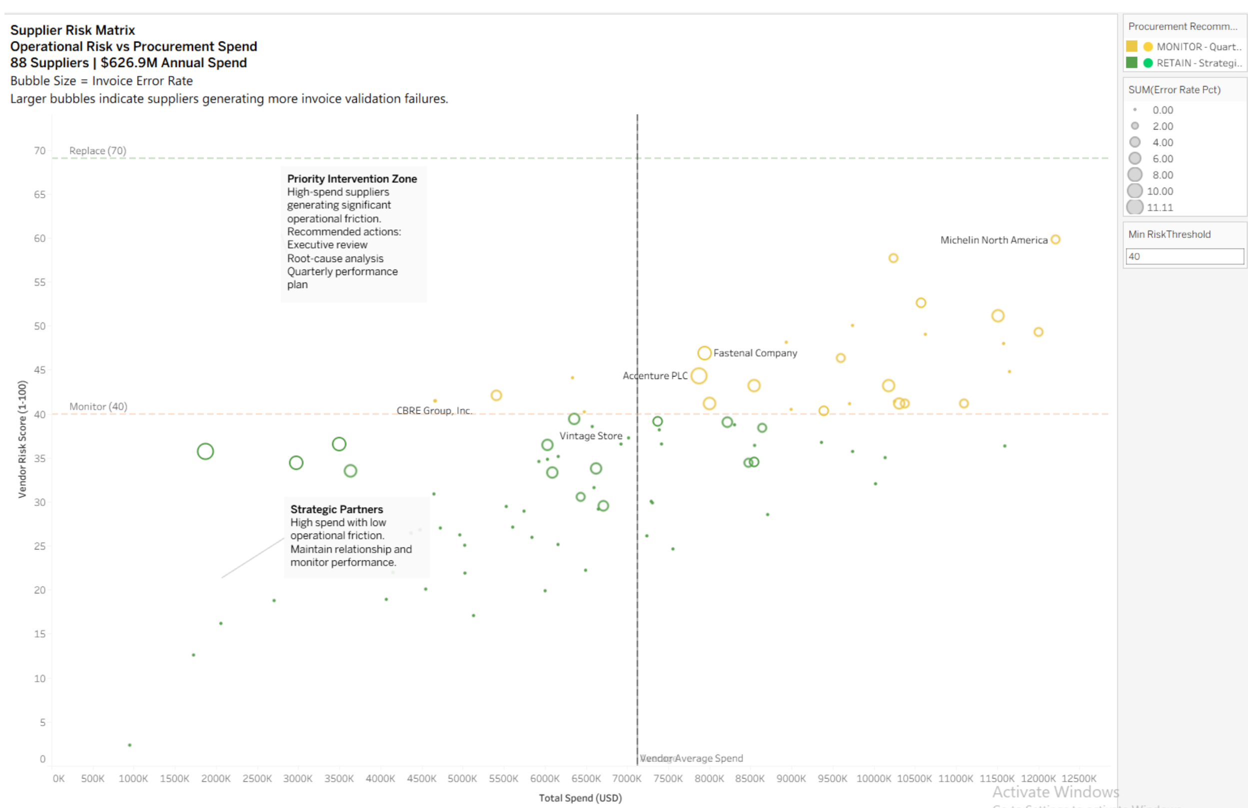
Meanwhile, the **bottom-right quadrant** highlights your **strategic partners**: high spend, but low risk. These are the suppliers you can rely on, the ones that keep your operations smooth without the headaches.

But how do you decide what to do? The chart makes it easy with **decision thresholds**. The reference lines at **Risk Score 40 (Monitor)** and **70 (Replace)** act as clear signposts, telling you exactly when to step in and when to step back.

And then there's the **operational friction**—the hidden cost of doing business. Here, it's visualized through bubble size: the larger the bubble, the higher the error rate, and the more friction that supplier is adding to your operations.

Of course, not all suppliers are created equal. That's why we've **annotated the exceptions**—the standout vendors like Michelin, Accenture, CBRE, Fastenal, and Vintage Store. Their stories are the ones that matter most.

Finally, there's a silver lining: **portfolio stability**. No suppliers currently sit above the Replace threshold of 70. For now, the risk is manageable—but the chart keeps you prepared for whatever comes next.



Key Takeaways

From this chart, executives can immediately see:

1. No suppliers exceed the Replace threshold (70) —the portfolio is not in crisis.
2. 18 suppliers fall into the Monitor zone (40–69) —representing $174.8M (28% of spend) requiring quarterly performance reviews.
3. Michelin combines the highest spend ($12.2M) with a 25% rejection rate—the highest-priority monitoring target.
4. Accenture has the highest invoice error rate (10.71%) despite being classified as "Low Risk"—the legacy risk label is misleading.
5. CBRE has the highest payment rejection rate (56.3%)—an immediate AP process review is required.
6. Vintage Store, the sole "High Risk" vendor, performs flawlessly (0% errors)—demonstrating that legacy risk labels are unreliable.

The Supplier Risk Matrix is the centerpiece of the Procurement Intelligence Platform. It answers the executive question: 'Which suppliers should I focus on?' by combining financial exposure, operational risk, and decision thresholds into a single, intuitive visualization. Every design decision—color, size, reference lines, annotations—was made to support a specific executive decision.

Chart 2: Spend Exposure by Supplier Risk

Introduction

Purpose of This Chart

The treemap serves a distinct purpose from the scatter plot in Chart 1. While the scatter plot helps procurement leaders identify which individual suppliers require attention, the treemap answers a higher-level strategic question: *"Where is our money concentrated, and how much of it is exposed to risk?"*

This distinction is critical. A procurement executive needs to understand both:

1. Which suppliers to focus on (Chart 1)
2. How much of the total portfolio is at risk (Chart 2)

The treemap is the ideal visualization for this because it encodes two dimensions simultaneously:

- Area represents financial exposure (larger blocks = more spend)
- Color represents risk status (Green = Retain, Amber = Monitor, Red = Replace)

This dual encoding allows an executive to scan the entire $627M portfolio in seconds and immediately answer the question: *"How much of our spend is safe, and how much requires oversight?"*

Why We Chose a Treemap

When it comes to understanding spend, proportional representation is everything. That's why we turned to a **treemap**. Here, **block size equals spend**, so the largest blocks immediately reveal where the money is going. No other chart communicates relative proportion with such clarity—what you see is what you get.

But spend alone doesn't tell the full story. By overlaying **color**, we add risk status into the mix. Now, executives can instantly see not just *where* the money is going, but *how much of it is at risk*. Green for safe, amber for caution—it's all there at a glance.

The treemap also reflects the natural hierarchy of procurement. Large blocks dominate the view, while smaller ones fade into the background, mirroring how teams prioritize in the real world. The most consequential suppliers demand attention, while the rest provide context.

And then there's the **space efficiency**. With 88 suppliers displayed in a single view, there's no need for scrolling or filtering. A bar chart would force you to dig for insights, but the treemap lays it all out in one clean, intuitive view.

Perhaps most importantly, the treemap delivers **immediate insight**. The balance of green versus amber blocks answers the critical question—*How healthy is our portfolio?*—instantly. No analysis required. The answer is right there in the colors.

What This Chart Reveals

This treemap cuts straight to the heart of the executive question: *"Which suppliers represent our greatest financial exposure, and where should we focus our oversight?"*

Here's how it delivers the answer:

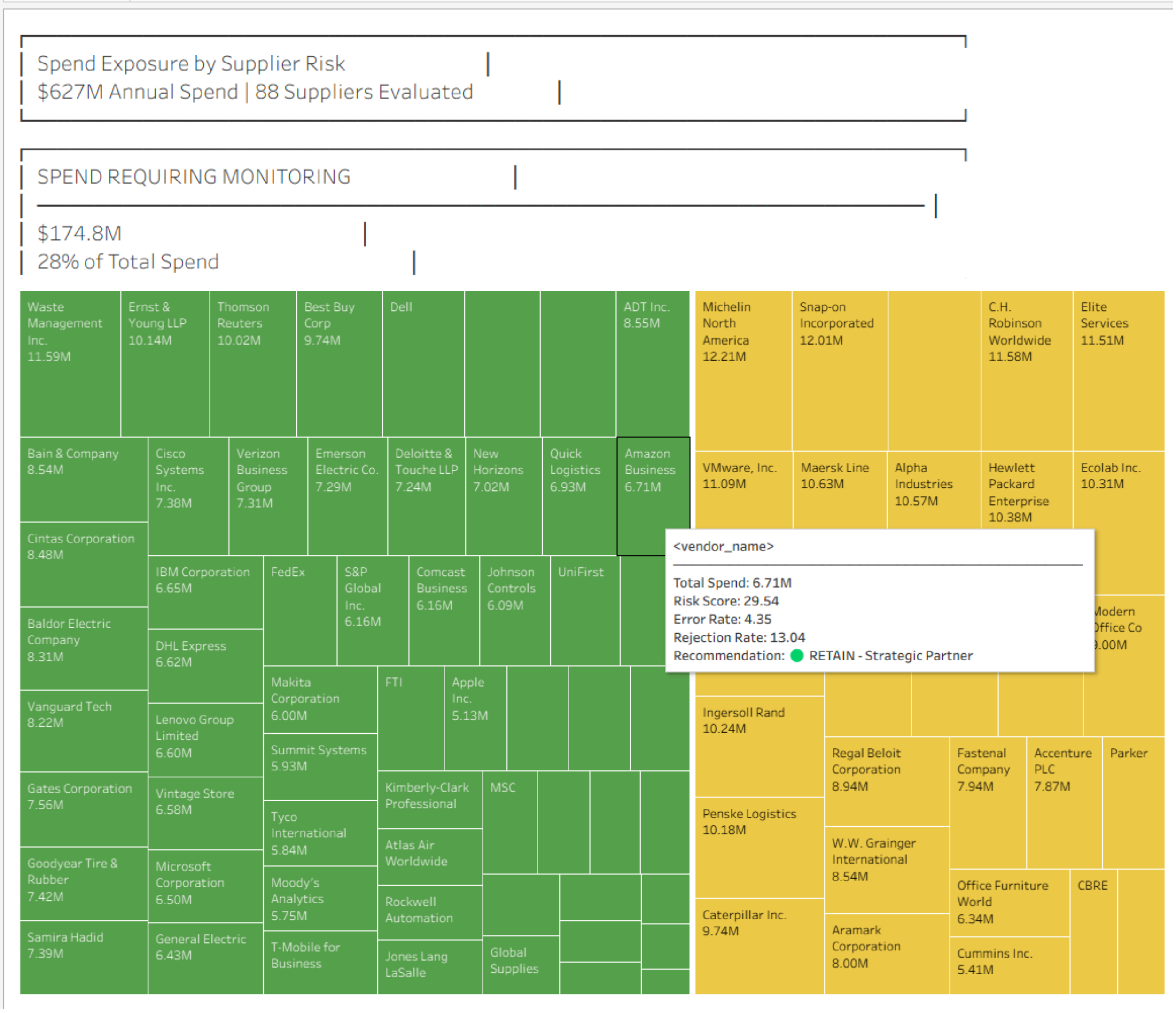
First, **spend concentration** is impossible to miss. The largest blocks dominate the view, instantly showing you where your money is going. No calculations, no guesswork—just clarity.

Then, there's **risk distribution**. The color coding tells the story at a glance: **green for Retain**, **amber for Monitor**, and **red for Replace**. You don't just see the spend; you see the risk attached to it.

And that leads to the bigger picture—**portfolio health**. The balance of green versus amber blocks gives you an immediate sense of stability. Are you in good shape, or is it time to act? The answer is right there in the colors.

Finally, the treemap points you to **action priority**. The largest amber blocks? Those are your highest-priority monitoring targets. They're the suppliers that demand your attention *now*—the ones where oversight can make the biggest difference.

In short, this chart doesn't just show data. It tells you where to look, what to worry about, and what to do next.



Key Takeaways

From this chart, executives can immediately see:

1. $174.8M (28%) of procurement spend falls within the Monitor zone—these suppliers require quarterly performance reviews.
2. The largest amber blocks—Michelin ($12.2M), Snap-on ($12.0M), and Waste Management ($11.6M) —represent the highest-priority monitoring targets.
3. No suppliers exceed the Replace threshold, indicating the portfolio is operationally stable.

4. 72% of spend resides with low-friction suppliers (green blocks), confirming the portfolio is healthy overall.
5. Supplier concentration is moderate—the top 10 suppliers account for 17.3% of total spend.

*The largest yellow blocks—Michelin ($12.2M), Snap-on ($12.0M), and Waste Management ($11.6M)—represent the Monitor cohort. Green blocks dominate, confirming most spend resides with low-friction suppliers.*s

Chart 3: Performance vs. Benchmarks

Introduction

Purpose of This Chart

The scatter plot (Chart 1) identifies *which suppliers deserve attention* based on overall risk. The treemap (Chart 2) shows *how much spend is exposed to risk*. This chart answers a different question: *"Why are those vendors risky?"*

This distinction matters. A supplier may have a moderate risk score but still generate significant operational drag through high error rates or payment rejections. This chart isolates these operational metrics and benchmarks them against portfolio averages, revealing:

- Which vendors exceed average error rates (invoice quality issues)
- Which vendors exceed average rejection rates (payment processing friction)
- Which vendors are double threats (exceeding both averages)

Why a Diverging Bar Chart?

A **Diverging Bar Chart**—also known as a Diverging Stacked Bar Chart or Bullet Chart—isn't just another way to display data. It's a tool designed for clarity and action.

At its core, this chart **compares values to a benchmark**, like an average rejection rate. But it doesn't stop there. It **highlights deviations**—whether above or below the benchmark—using color and direction. Red for trouble, green for success. The message is instant: *Are we above or below the line?*

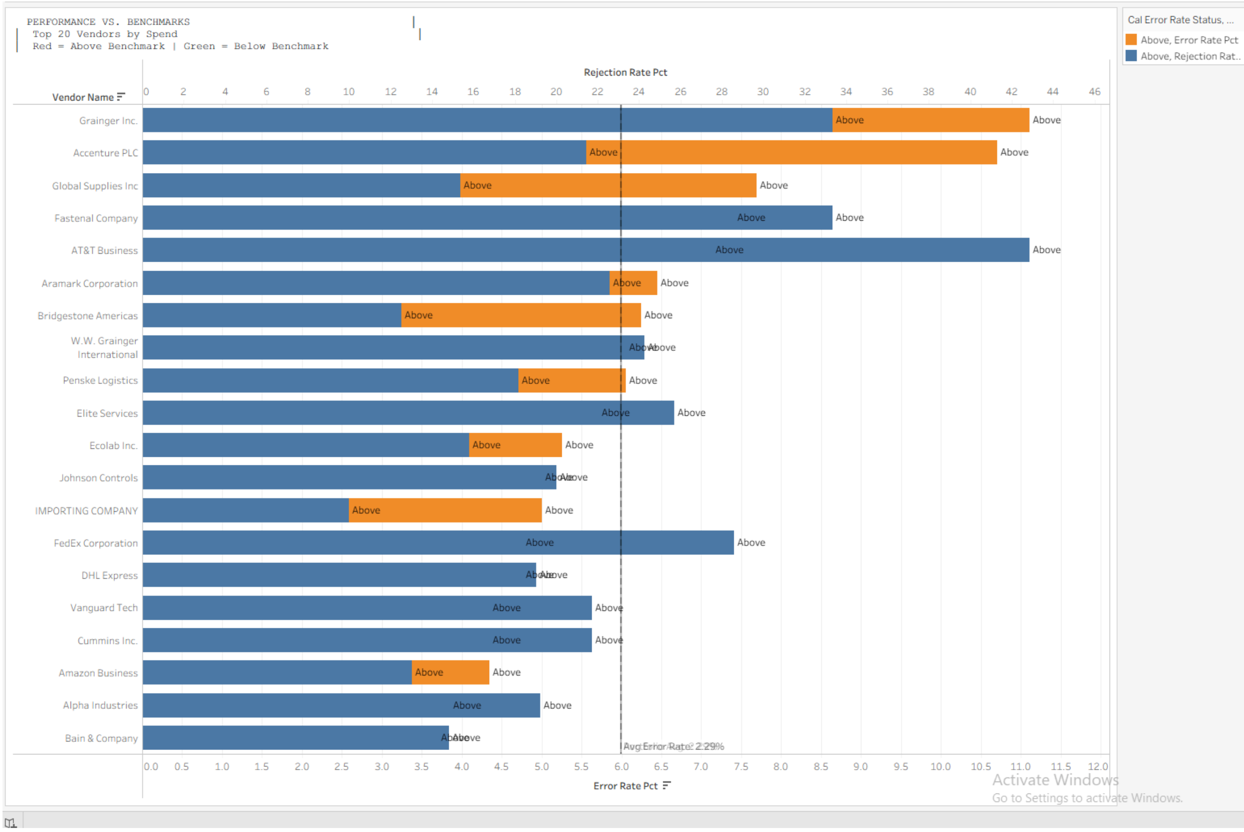
And because procurement isn't one-dimensional, this chart **displays multiple metrics side-by-side**. Error Rate. Rejection Rate. Whatever matters most. You see it all in one place, without the clutter.

But perhaps its greatest strength is **space efficiency**. Need to compare dozens of vendors? No problem. The diverging bar chart packs them all into a compact, scannable format. No scrolling, no flipping between screens—just the insights you need, at a glance.

For your procurement dashboard, this chart is a perfect fit. It answers the questions that keep your team up at night:

- **Which vendors are underperforming?** (The ones above the average rejection or error rates.)
- **Which vendors are exceeding expectations?** (The ones below the benchmark.)
- **How do vendors stack up across multiple metrics?** (Error Rate vs. Rejection Rate—side by side, no guesswork.)

In short, the diverging bar chart doesn't just show data. It tells you what's working, what's not, and where to focus next.

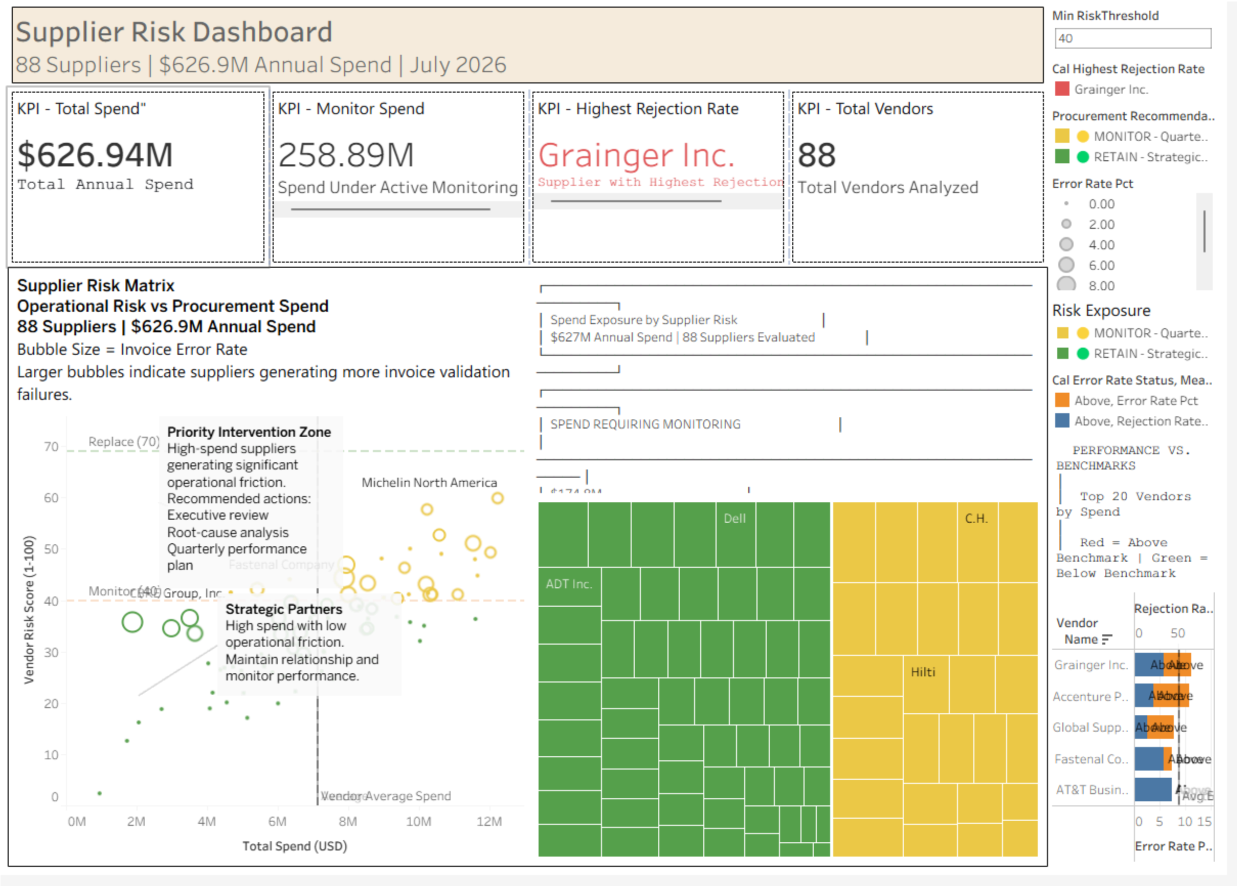


Accenture's error bar extends past the 2.29% line—a 4.7× deviation. CBRE's rejection bar nearly triples the 18.76% average. Fastenal shows both bars extended, confirming highest combined friction.

Chart 4: Supplier Risk Dashboard Overview

Executive Dashboard with KPI Cards

Purpose: This dashboard provides a **real-time, data-driven view** of supplier risk and spend across **88 suppliers**, totaling **$626.9M in annual spend** (as of July 2026). It empowers procurement teams to **prioritize oversight, mitigate risks, and optimize supplier relationships** through actionable insights.



[See on Tableau Public](#)

Key Components & Insights

1. **High-Level KPIs**

   - **Total Annual Spend: $626.94M** across all suppliers.
   - **Spend Under Active Monitoring: $258.89M** (suppliers flagged for closer oversight).
   - **Highest Rejection Rate: Grainger Inc.** (highlighted as the supplier with the most frequent rejections).
   - **Total Vendors Analyzed: 88 suppliers** evaluated for risk and performance.

2. **Supplier Risk Matrix (Scatter Plot)**

   - **Axes:**
     - **X-axis (Total Spend):** Financial exposure per supplier.
     - **Y-axis (Vendor Risk Score):** Operational risk (0–100 scale).

    - **Bubble Size:** Represents **invoice error rate—**larger bubbles indicate more validation failures.
    - **Zones:**
        - **Priority Intervention Zone (Risk Score ≥ 70):** High-spend, high-risk suppliers requiring **executive review, root-cause analysis, and quarterly performance plans**.
        - **Monitor Zone (Risk Score ≥ 40):** Suppliers with moderate risk; **active monitoring** recommended.
        - **Strategic Partners (Low Risk):** High-spend, low-risk suppliers; **maintain relationships and monitor performance**.
3. **Spend Exposure by Supplier Risk (Treemap)**

    - **Block Size:** Proportional to **spend—**larger blocks = higher financial exposure.
    - **Color Coding:**
        - **Green (Retain):** Suppliers performing well (e.g., Dell, ADT Inc.).
        - **Yellow (Monitor):** Suppliers requiring oversight (e.g., C.H., Hilti).
    - **Insight:** Visualizes **where money is concentrated** and **which suppliers pose the greatest risk**.
4. **Performance Benchmarks (Bar Chart)**

    - **Top 20 Vendors by Spend:** Compares **rejection rates** and **error rates** against benchmarks.
    - **Color Coding:**
        - **Red:** Above benchmark (underperforming).
        - **Green:** Below benchmark (exceeding expectations).
    - **Key Outlier: Grainger Inc.** has the highest rejection rate.
5. **Actionable Recommendations**

    - **Min Risk Threshold: 40** (suppliers below this are low-risk).
    - **Procurement Recommendations:**
        - **Retain (Green):** Strategic suppliers with low risk.
        - **Monitor (Yellow):** Quarterly reviews for moderate-risk suppliers.
        - **Replace (Red):** High-risk suppliers requiring intervention.

Why This Matters for Stakeholders

- **Cost Savings:** Identify high-spend, high-risk suppliers to **negotiate better terms or switch vendors**.
- **Risk Mitigation:** Proactively address suppliers with **high rejection/error rates** to avoid operational disruptions.
- **Efficiency:** Focus resources on **priority suppliers** (e.g., those in the "Replace" or "Monitor" zones).
- **Strategic Decision-Making:** Use the **treemap and scatter plot** to align procurement strategy with financial and operational goals.

Next Steps for Discussion:

- Should we **prioritize renegotiations** with high-risk, high-spend suppliers like Grainger Inc.?
- How can we **reduce invoice errors** for suppliers with large bubbles in the scatter plot?
- Are there **opportunities to consolidate spend** with strategic partners (green zone)?

---

6. Assumptions & Limitations

This section acknowledges the boundaries of the analysis. It demonstrates analytical discipline and prevents overclaiming.

This analysis is based solely on ERP vendor master data and invoice transaction history. It does not incorporate:

- Supplier financial health, credit ratings, or bankruptcy risk.
- Delivery performance, lead times, or quality of goods/services received.
- Contract compliance, negotiated discounts, or pricing benchmarks.
- Inventory impacts, safety stock, or supply chain substitution costs.
- External macroeconomic or geopolitical risk factors.

Consequently, the Vendor Risk Score is an operational procurement friction indicator designed to prioritize AP process improvements and vendor performance management, not a comprehensive enterprise-wide supplier risk assessment. All findings flagged as "investigation warranted" are empirically-grounded hypotheses that require domain-specific validation by Procurement and AP teams.

---

7. Actionable Recommendations

This section translates the analytical findings into specific procurement actions. Each recommendation includes the owner, timeline, and business rationale.

Immediate Actions (Next 30 Days)

Action	Owner	Rationale
Investigate CBRE Group remittance process	AP Manager	56.25% rejection rate—highest in dataset. Validate bank details, PO references, and approval workflows.
Audit Accenture's invoice template and PO matching	AP Lead + Vendor Manager	10.71% error rate—4.7× above average. Review tax IDs, billing codes, and formatting requirements.

Schedule performance review with Fastenal	Procurement Director	7.41% errors + 33.33% rejections. Initiate a structured improvement plan.
Validate payment terms with Michelin	AP Specialist	25% rejection rate with low errors—verify Net-term alignment (e.g., Net 30 vs Net 60).

Short-Term Actions (90 Days)

- Implement Quarterly Business Reviews (QBRs) for all 18 vendors scoring 40–69. Track Error and Rejection rates on a shared dashboard.
- Establish internal alerting thresholds:
  - If any vendor exceeds 10% rejection rate, trigger an internal process audit.
  - If any vendor exceeds 5% error rate, request corrected invoice templates.

Long-Term Strategy (12 Months)

- Re-run this scorecard quarterly. The Monitor cohort (currently 18 vendors) should shrink as performance improves.
- Apply this scoring framework to all new vendor onboarding to establish baseline expectations for invoice quality and payment reliability.

---

8. Appendix: SQL Code Used

This section provides the technical foundation for the analysis. The full SQL query is provided below to ensure reproducibility and transparency.

```sql
WITH cte_vendor_metrics AS (

    SELECT

        v.vendor_id,

        v.vendor_name,

        v.risk_level,

        SUM(i.total) AS total_spend,

        COUNT(i.invoice_id) AS invoice_count,

        SUM(CASE WHEN i.validation_status = 'Rejected' THEN 1 ELSE 0 END) AS
failed_validations,

        SUM(CASE WHEN i.status = 'Rejected' THEN 1 ELSE 0 END) AS
rejected_payments,

        ROUND(100.0 * SUM(CASE WHEN i.validation_status = 'Rejected' THEN 1
ELSE 0 END) / NULLIF(COUNT(i.invoice_id), 0), 2) AS error_rate_pct,

        ROUND(100.0 * SUM(CASE WHEN i.status = 'Rejected' THEN 1 ELSE 0 END) /
NULLIF(COUNT(i.invoice_id), 0), 2) AS rejection_rate_pct

    FROM vendors_july_2026 v

    JOIN invoice_july_2026 i ON v.vendor_id = i.vendor_id

    WHERE v.currency = 'USD'

    GROUP BY v.vendor_id, v.vendor_name, v.risk_level

),

scored_vendors AS (

    SELECT

        vendor_name,
```

```
        total_spend,

        invoice_count,

        risk_level,

        error_rate_pct,

        rejection_rate_pct,

        ROUND((total_spend / (SELECT MAX(total_spend) FROM
cte_vendor_metrics)) * 30, 2) AS spend_score,

        CASE

            WHEN error_rate_pct >= 30 THEN 30

            ELSE ROUND((error_rate_pct / 30) * 30, 2)

        END AS error_score,

        CASE

            WHEN rejection_rate_pct >= 30 THEN 20

            ELSE ROUND((rejection_rate_pct / 30) * 20, 2)

        END AS rejection_score,

        CASE risk_level

            WHEN 'High' THEN 20

            WHEN 'Medium' THEN 10

            WHEN 'Low' THEN 0

            ELSE 0

        END AS risk_score

    FROM cte_vendor_metrics

)

SELECT
```

```
    vendor_name,

    total_spend,

    invoice_count,

    risk_level,

    error_rate_pct,

    rejection_rate_pct,

    ROUND((spend_score + error_score + rejection_score + risk_score), 2) AS
vendor_risk_score,

    CASE

        WHEN (spend_score + error_score + rejection_score + risk_score) >= 70
THEN '🔴 REPLACE - Phase Out in 6 Months'

        WHEN (spend_score + error_score + rejection_score + risk_score)
BETWEEN 40 AND 69 THEN '🟡 MONITOR - Quarterly Review Required'

        ELSE '🟢 RETAIN - Strategic Partner'

    END AS procurement_recommendation,

    CASE

        WHEN (spend_score + error_score + rejection_score + risk_score) >= 70
THEN

            CONCAT('Score ', (spend_score + error_score + rejection_score +
risk_score), '/100 - Prioritize replacement')

        WHEN total_spend = (SELECT MAX(total_spend) FROM cte_vendor_metrics)
THEN

            CONCAT('⭐ HERO FINDING: Highest spend vendor ($',
ROUND(total_spend/1000000, 1), 'M) - Monitor closely')

        ELSE

            CONCAT('Score ', (spend_score + error_score + rejection_score +
risk_score), '/100 - Standard oversight')
```

```sql
    END AS executive_insight
FROM scored_vendors
ORDER BY (spend_score + error_score + rejection_score + risk_score) DESC;
```

---

9. Project Summary & Extension

This section concludes the project and positions it within the broader Procurement Intelligence Platform.

This report concludes Phase 1 of the Procurement Intelligence Platform initiative. The analysis successfully transitions vendor oversight from static, subjective risk labels to a dynamic, transaction-based scoring model.

Key project outcome: The scorecard identified 18 vendors ($175M in spend) requiring active monitoring—a 28% spend cohort that Procurement can now manage proactively through quarterly reviews and structured performance improvement plans.

Next phases in the initiative:

- Phase 2: Tableau dashboard implementation (as outlined in Section 5) for ongoing self-service monitoring by the Procurement team.
- Phase 3: Expansion to non-USD vendors and integration of delivery performance metrics.
- Phase 4: Development of automated alerting triggers within the ERP.

Case Study 2: Strategic Contract Optimization

Project Phase: 1 – Data Exploration & Decision Engine Development
Date: July 2026
Author: Ebhota Oje
Distribution: Chief Financial Officer (CFO), Procurement Director, Category Managers

---

1. Executive Summary

This section provides a high-level overview of the business problem, analytical approach, key findings, and the most critical insight for procurement and finance leadership. It is designed to be read in under two minutes.

Block	Summary
Problem	Thousands of products purchased transactionally. No volume discounts. High admin overhead.
Approach	Weighted scorecard: Spend 30% + Frequency 25% + Consistency 20% + Volatility 15% + Concentration 10%.

Outcome	1 product (Power Tools) >70. 50+ in review. 35+ phase-out. $2.7M savings opportunity.
Recommendation	Negotiate Power Tools contract. Phase out Implementation Service (-93.8%).

🟢 Hero Finding

Traditional aggregate analysis would have recommended contracts for dying products.

Observation: Implementation Service had $2.69M in 2023 spend but **$11K in 2025** (-93.8% decline).

Evidence: Trend overlay triggered Phase-Out Override (YoY < -50% AND 2025 spend < 25% of 2023 spend).

Investigation: Review supplier relationship for Implementation Service. Consider exit or renegotiation.

The Business Problem

Procurement purchases thousands of products across dozens of categories. Currently, most purchasing happens transactionally—meaning:

- No volume discounts – suppliers have no incentive to offer tiered pricing without annual commitments.
- Inconsistent pricing – spot purchases are subject to market fluctuations.
- Fragmented procurement – different departments buying the same items from different vendors.

- High administrative overhead – frequent purchase orders increase procurement effort, invoice processing, approvals, and supplier management costs even when individual order values are relatively small.

The Procurement Director needs to know: *"Which products should we commit to annual contracts, and which should remain on-demand purchases?"*

The Analytical Approach

Using ERP data spanning 2023–2025, we developed a weighted procurement decision engine that evaluates products across five strategic dimensions:

Dimension	Weight	Rationale
Spend Impact	30%	Financial materiality—high spend justifies negotiation effort
Purchase Frequency	25%	Administrative overhead reduction—frequent POs create operational waste
Demand Predictability	20%	Consistent demand makes long-term commitments safe
Price Volatility Opportunity	15%	High volatility creates value in locking fixed pricing

Supplier Concentration	10%	Single/dual-source suppliers carry supply risk that contracts mitigate

The Outcome

- 1 product (Power Tools) scored ≥70 and is recommended for immediate contract negotiation, representing $2.47M in annual spend** and **~$198K in estimated annual savings.
- 50+ products scored 40–69 and are flagged for strategic review—these require business validation but represent the next wave of contracting opportunity.
- 35+ products were automatically flagged for phase-out due to demand collapse (YoY decline >50% AND 2025 spend <25% of 2023 spend). These products—including Implementation Service (-93.8%), Strategic Consulting (-82.1%), and Facility Maintenance (-81.8%)—are NOT suitable for contracts and would have been erroneously recommended by aggregate-only models.

The Hero Finding: Traditional aggregate analysis would have flagged Implementation Service and Consulting Services as "strong contract candidates" based on historical spend. Our trend overlay prevented this error, protecting procurement from signing long-term agreements for products that are effectively dying.

Estimated Total Addressable Savings: Across the top 20 contract candidates, the model identifies ~$2.7M in annual savings opportunity through strategic contracting and pricing agreements.

---

2. Methodology

This section explains the analytical framework in detail. Each design decision is accompanied by its business rationale so that the methodology is defensible and *transparent. Key areas covered include data sources, metric definitions, volatility calculations, and the critical phase-out override.*

2.1 Data Sources & Scope

The analysis is based on three years of ERP transaction data (2023–2025), filtered to products sourced from vendors with USD as their primary currency.

Source	Table	Description
Invoice Line Items	`invoice_line_items_july_2026`	Product-level transaction data
Invoice Headers	`invoice_july_2026`	Vendor, date, and PO references
Vendor Master	`vendors_july_2026`	Vendor currency and classification
Scope Filter	`v.currency = 'USD'`	Aligns with US-dollar reporting base
Time Period	2023–2025	Three-year trend analysis for demand stability

2.2 Core Metrics Calculated

Each metric is designed to answer a specific business question about contract suitability. The metrics are intentionally diverse to capture different dimensions of procurement value.

Metric	Definition	Business Relevance
Annual Spend	Average annual spend across 2023–2025	Financial leverage for negotiation
Purchase Frequency	Average number of POs per year	Administrative overhead indicator
Demand Consistency	Average months with purchases (out of 12)	Demand predictability—safer to contract
Price Volatility (CoV)	Coefficient of Variation = (STDDEV(unit_price) / AVG(unit_price)) × 100	Price stability. Lower CV = more predictable costs

Supplier Concentration	Number of unique suppliers per product	Supply risk—fewer suppliers = higher dependency

2.3 The Volatility Calculation (Why We Changed It)

This section explains a critical methodological improvement. The original range-based volatility metric was sensitive to outliers; the revised Coefficient of Variation (CoV) approach provides a more accurate and defensible measure of price stability.

The Original (Flawed) Approach

The initial model used:

```text
Price Volatility = (MAX(unit_price) - MIN(unit_price)) / AVG(unit_price) × 100
```

Why this was problematic: This "range-based" metric is highly sensitive to outliers—a single anomalous invoice can completely distort the score.

Example:

Month	Price
Jan–Nov	$100, $105, $102, $108, $101, $103, $107, $104, $102, $106, $103

Dec	$500 (one-off rush order)

-
  Range-based volatility: (500 - 100) / 105 × 100 = 381% → This product appears "highly volatile" despite 11 months of stable pricing.
- CoV-based volatility: STDDEV(95) / AVG(105) × 100 = ~12% → This accurately reflects the 11 months of stability and flags the December order as an outlier.

The Fix – Coefficient of Variation (CoV):

```text
CoV = (STDDEV_POP(unit_price) / AVG(unit_price)) × 100
```

CoV is a standardized, scale-independent measure of dispersion. It tells procurement: *"On average, unit prices deviate by X% from the mean."*

CoV Range	**Interpretation**	**Recommended Action**
CoV < 20%	Stable pricing	Safe to contract
CoV 20–50%	Moderate volatility	Consider fixed-price contract
CoV > 50%	Highly volatile	Consider index-based or flexible pricing

Scoring Logic: Since we want stable pricing to be rewarded (safe contracts), the score is inverted:

```text
Volatility Score = (1 - (Product_CoV / Max_CoV_Across_All_Products)) × 15
```

- If a product has the lowest CoV (most stable), it gets the full 15 points.
- If a product has the highest CoV (most volatile), it gets 0 points.

2.4 Supplier Concentration Scoring

This dimension recognizes that procurement's role includes supply continuity, not just cost reduction. A single-source product with stable demand is a higher priority for contracting than a commodity available from dozens of vendors.

```text
Concentration Score = (1 - (Product_Supplier_Count /
Max_Supplier_Count_Across_All_Products)) × 10
```

Supplier Count	Score	Business Implication
1 supplier	10	High supply risk → Strong contract incentive to secure capacity
10 suppliers	0	Low supply risk → Less urgent to contract

2.5 Phase-Out Override (The "Guardrail")

A critical insight from the trend analysis: "High historical spend does not guarantee future demand." This override prevents procurement from signing contracts for products that are effectively dying.

Without this override, Implementation Service would have been considered a "Candidate for Review" based on its average annual spend ($961K) and moderate frequency. But looking at the trend:

Year	**Spend**
2023	$2,689,478
2024	$182,814
2025	$11,330

YoY Decline: -93.8%

The Phase-Out Override triggers when:

```text
YoY Growth < -50%

AND

Spend_2025 < Spend_2023 × 0.25
```

If this condition is met, the product is automatically downgraded to 🔻 PHASE-OUT, regardless of the contract priority score.

Business Rationale: Signing a long-term contract for a product whose demand has collapsed by 93.8% would be a catastrophic waste of procurement resources and budget. This override acts as a critical safeguard against inertia-based decision-making.

2.6 Low-Base Growth Signal (The "Context Flag")

The model also flags products where high growth is mathematically impressive but operationally meaningless due to a low starting base. This prevents overreaction to statistical anomalies.

Example: Air Tools

Year	Spend
2023	$2,159,199
2024	$76,360
2025	$1,411,596

YoY Growth: +1748%

Mathematically, this is spectacular growth. But 2024's spend ($76K) is only 6% of the average annual spend ($1.2M). The `growth_signal_quality` flag applies:

```text
IF spend_2024 < avg_annual_spend × 0.25 AND yoy_growth_pct > 20

THEN '⚠️ Low Base Growth (Context Required)'
```

This doesn't reject the product—it simply tells the Category Manager: *"This growth is phenomenal, but please validate the 2024 base before relying on it."*

2.7 The Weighted Score Calculation

For each product, the final Contract Priority Score combines all five dimensions into a single 0–100 metric. The weights reflect the relative importance of each dimension to procurement decision-making.

```text
Score =
    (Avg_Spend / Max_Spend) × 30          [Spend Impact]
    + (Avg_Frequency / Max_Frequency) × 25  [Frequency]
    + (Avg_Months / 12) × 20               [Consistency]
    + (1 - Avg_CV / Max_CV) × 15            [Volatility - Inverted]
    + (1 - Avg_Suppliers / Max_Suppliers) × 10 [Concentration - Inverted]
```

Example: Power Tools

Metric	Value	Calculation	Points
Spend	$2,473,668	($2.47M / $2.47M) × 30	30.00
Frequency	10 POs/year	(10 / 10) × 25	25.00
Consistency	9 months	(9 / 12) × 20	15.00

Volatility	CoV = 100.2%	(1 - 100.2 / 137.1) × 15	4.38
Concentration	3 suppliers	(1 - 3 / 3) × 10	0.00
Total			74.38

Why Power Tools scores zero on concentration: It has 3 suppliers—the highest count in the dataset. There is no urgency to contract for supply security; the urgency comes purely from spend and frequency.

2.8 Decision Thresholds

The thresholds translate the numerical score into clear procurement actions. The 70-point threshold ensures that only products with a compelling business case proceed to contract negotiation.

Score Range	Classification	Action
≥70	🔴 Strong Contract Candidate	Prioritize for immediate contract negotiation

40–69	🟡 Strategic Review	Validate with business stakeholders; likely contract candidates
<40	🟢 Maintain Flexible Purchasing	Continue on-demand or blanket PO; contract not justified
Phase-Out Override	🔻 Phase-Out / Demand Collapsed	Do NOT contract; review supplier relationship

2.9 Contract Type Recommendations

Based on product category classification, the model suggests a specific contract structure. This ensures the contract type matches the commercial characteristics of the product category.

Category Signal	Contract Type	Business Rationale
Consulting, Advisory, Audit, Legal	📄 Rate Card / SOW	Fixed rates for defined services; flexible scope

Freight, Shipping, Cargo, Logistics	📊 Index-Based Pricing	Link to fuel or market indices; protects against market spikes
License, Software, Subscription, Cloud	💻 Enterprise License Agreement	Volume-based licensing; predictable annual cost
Tire, Truck, Fleet	🔧 Volume Agreement (Tiered)	Tiered discounts based on volume tiers
Hardware, Tools, Parts (Default)	📦 Annual Volume Contract	Fixed pricing for annual volume commitment

---

3. Key Findings

This section presents the results of the analysis. Each finding is presented with supporting data and interpreted through the lens of the scoring model. The findings are organized into three categories: top contract candidates, phase-out warnings, and savings opportunities.

3.1 Top Contract Candidates

The following products scored 50+ and demonstrate strong growth trends. These are the highest-confidence contracting opportunities in the portfolio.

Rank	Product	Avg Spend	YoY Growth	Score	Savings Opportunity	Contract Type
1	Power Tools	$2.47 M	+130%	74.38	$197,893	📦 Volume Contract
2	Spare Parts	$1.93 M	+85%	59.38	$173,827	📦 Volume Contract
3	Truck Tires	$1.78 M	+90%	53.68	$160,310	🔧 Volume Agreement
4	Hardware	$1.58 M	+127%	51.71	$141,846	📦 Volume Contract
5	Fasteners	$1.42 M	+146%	51.10	$127,777	📦 Volume Contract

Key Insight: All top 5 candidates show organic growth >85%, confirming that demand is genuinely increasing. This is not a "low-base anomaly"—these are high-confidence contracting opportunities.

3.2 The Phase-Out List (Critical Warning)

The following products have experienced catastrophic demand collapse and should NOT be contracted. This is the most critical finding of the analysis—without the trend overlay, these products would have been erroneously recommended for contracts.

Product	2023 Spend	2025 Spend	Decline	Recommendation
Implementation Service	$2.69M	$11,330	-93.8%	🔻 Phase-Out
Strategic Consulting	$1.72M	$218,785	-82.1%	🔻 Phase-Out
Facility Maintenance	$1.56M	$233,956	-81.8%	🔻 Phase-Out
Software License	$2.30M	$16,841	-90.3%	🔻 Phase-Out
Market Analysis	$1.87M	$232,972	-75.7%	🔻 Phase-Out
Consulting Services	$1.70M	$467,075	-69.4%	🔻 Phase-Out

Executive Takeaway: *"Had we used the original aggregate model, Consulting Services ($1.7M historical average) would have been a top-tier contract candidate. The trend overlay prevented this error—demand has collapsed by 69%."*

3.3 Savings Opportunity Heatmap

This heatmap identifies the highest-ROI contracting opportunities based on estimated annual savings. These products should be prioritized for negotiation.

Product	Annual Spend	Estimated Savings	Action
Power Tools	$2.47M	$197,893	✅ Contract
Spare Parts	$1.93M	$173,827	✅ Contract
Truck Tires	$1.78M	$160,310	✅ Contract
Hardware	$1.58M	$141,846	✅ Contract
Fleet Services	$1.60M	$144,077	✅ Contract

Storage Array	$1.57M	$141,125	✅ Contract

Total Top 6 Savings: ~$959,078 annually.

---

4. Visualizations Blueprint (Phase 2 – Dashboard Design)

This section provides a blueprint for the Tableau dashboard that will visualize the contract optimization model. Each chart is designed to answer a specific executive question and support data-driven decision-making.

Chart 1: Strategic Contract Matrix (Revised)

Introduction

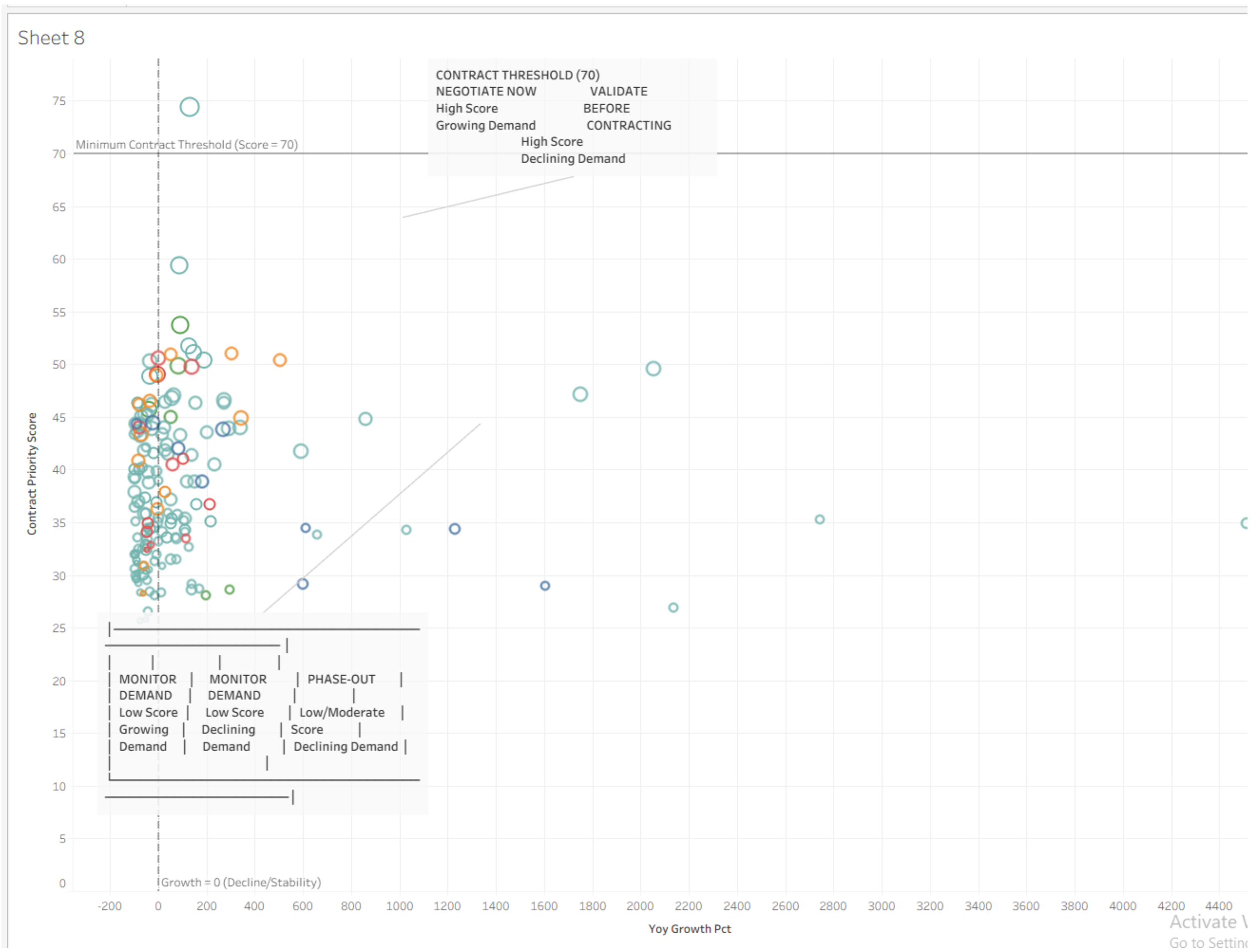
Purpose of This Chart

The Strategic Contract Matrix is the centerpiece of the Contract Intelligence dashboard. While the overall case study answers *"Which products should we put under contract?"*, this chart answers the immediate executive question: *"Which products should we negotiate contracts for immediately?"*

A procurement executive needs to identify products that combine high contract attractiveness with positive demand momentum. The scatter plot is the ideal visualization for this because it plots products along two critical dimensions simultaneously:

- X-Axis (Annual Demand Growth) represents demand momentum—is this product growing or declining?
- Y-Axis (Contract Attractiveness Score) represents contract suitability—how strong is the business case for contracting?

This dual-axis encoding immediately reveals which products sit in the Negotiate Now zone—the products that should be prioritized for immediate contract negotiation.



Why a Scatter Plot Was Selected

A Tool for Balanced Decision-Making

Contract decisions aren't one-dimensional—they require balancing **suitability (score)** and **momentum (growth)**. A scatter plot brings this trade-off to life, making it easy to see which products excel in both areas and which fall short.

Clear Boundaries for Action

The chart includes **reference lines** at **Score = 70 (Minimum Contract Threshold)** and **Growth = 0 (Decline/Stability)**. These lines create distinct zones, so you instantly know whether a product is a candidate for negotiation, validation, or phase-out.

A Third Dimension: Financial Impact

Bubble size represents the **Estimated Annual Savings Opportunity**, adding depth to the analysis. Larger bubbles highlight products that not only score well but also offer the **highest financial upside**—helping you prioritize where to focus negotiations.

Color-Coded for Quick Decisions
Colors—**Blue (Negotiate), Amber (Validate), Green (Monitor), and Red (Phase-Out)**—provide immediate visual cues. No need to dig into the data; the chart tells you exactly what action to take.

Focused on What Matters
Only the most critical products are labeled, mirroring how executives review data: **scanning for exceptions and outliers** rather than getting lost in the details.

Natural Groupings for Strategic Clarity
The scatter plot divides products into **four intuitive quadrants**, each with clear business implications. Whether it's a high-potential star or a declining underperformer, the chart makes it obvious where each product stands.

What This Chart Reveals

Answering the Critical Question
This scatter plot directly addresses the executive priority: *"Which products should we prioritize for immediate contract negotiation?"*

How It Delivers Insights

- **Immediate Contract Candidates:** Found in the **top-right quadrant** (High Score + Positive Growth)—these are your best opportunities.
- **Phase-Out Candidates:** Located in the **bottom-left or bottom-right quadrants** with negative growth—products that may no longer be viable.
- **Decision Thresholds:** The **Score = 70** and **Growth = 0** reference lines act as clear decision points, separating high-potential products from the rest.
- **Financial Impact:** Bubble size reveals the **Estimated Annual Savings**, so you can quickly identify which products offer the **biggest financial rewards**.
- **Portfolio Health:** The distribution of products across the quadrants gives you a **snapshot of your contract readiness**—are most products ready for negotiation, or do they need further validation?

Chart 2: Savings Concentration Pareto Chart

Introduction

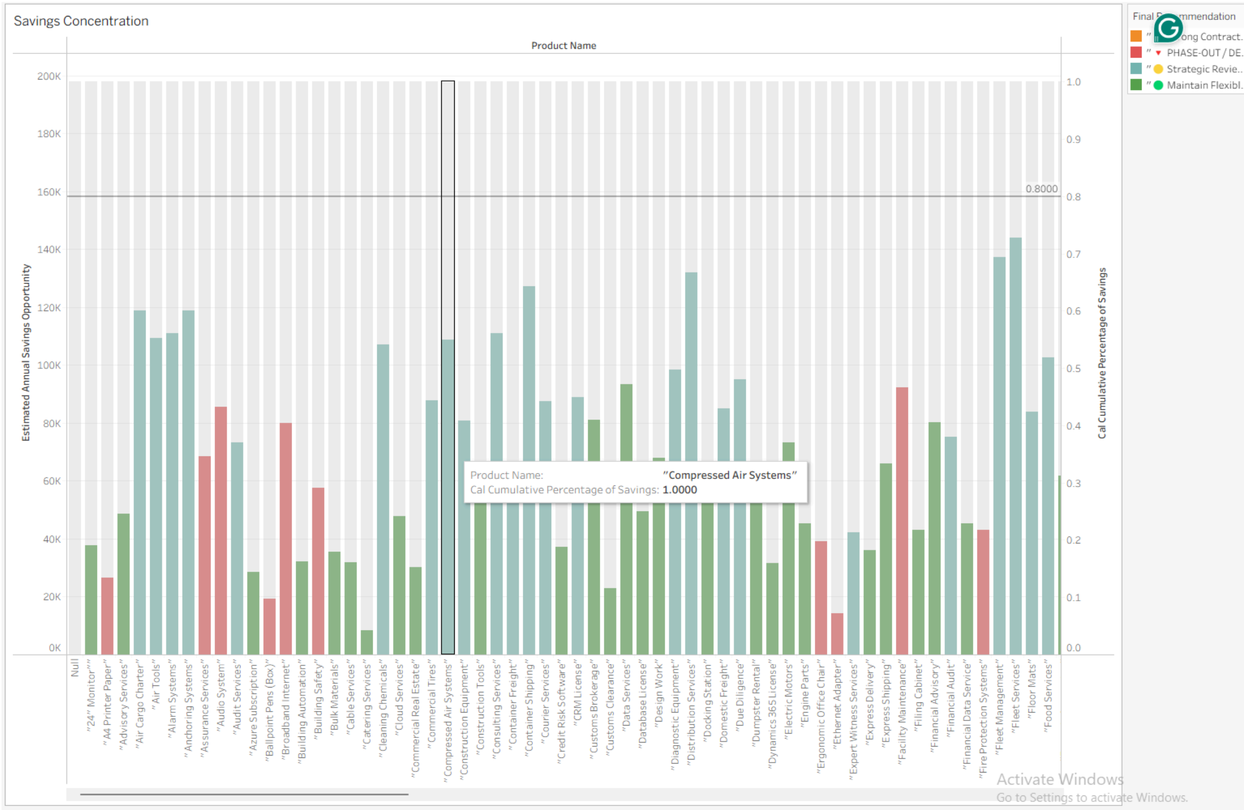
Purpose of This Chart

The scatter plot (Chart 1) identifies *which products to prioritize* based on contract attractiveness and demand growth. This chart answers the next critical question: *"How much value can we realistically capture, and where is it concentrated?"*

While Chart 1 tells procurement *which products to focus on*, Chart 2 quantifies the *concentration of value*. A Pareto chart is the ideal visualization because it immediately reveals:

- Which products generate the most savings (ranked bars)
- How many products generate 80% of the value (cumulative line)
- How concentrated the opportunity is (concentration ratio)

This is the question a Chief Procurement Officer actually asks: *"If I focus on the top X products, how much of the savings do I capture?"*



Why a Pareto Chart Was Selected

Uncovering the Vital Few

The Pareto Chart was chosen because it **instantly reveals concentration**—how a small number of products can generate the majority of savings. The **cumulative line** makes it clear: *Exactly how many products contribute to 80% of the savings?* No other chart answers this question so directly.

Designed for Executives

This chart speaks the language of leadership. It delivers a **clear, actionable message**: *"The top X products drive Y% of savings."* No complex analysis—just the insights leaders need to make quick, informed decisions.

Built for Decision-Making

Procurement teams often ask: *"How many products should we focus on?"* The Pareto Chart provides the answer. If **5 products account for 60% of savings**, the team knows exactly where to direct their efforts for maximum impact.

The Perfect Companion to the Scatter Plot

While **Chart 1 (Scatter Plot)** identifies *which* products to prioritize based on score and growth, **Chart 2 (Pareto Chart)** reveals *how much value* is concentrated in those top performers. Together, they create a complete picture: **direction (what to focus on) and scale (how much it matters).**

What This Chart Reveals

Answering the Big Question
This Pareto Chart directly addresses the executive priority: *"How many products generate the majority of savings, and where is the value concentrated?"*

How It Delivers Insights

- **Savings Concentration:** The **cumulative line** shows how many products contribute to 80% of the savings—helping you identify the "vital few."
- **Top Contributors:** The **ranked bars** highlight which products deliver the highest savings, so you know exactly where the biggest opportunities lie.
- **Diminishing Returns:** The **flattening cumulative line** signals when adding more products yields less value—helping you avoid spreading resources too thin.
- **Focus Area:** The **"80% threshold"** tells procurement precisely how many products to prioritize for maximum impact.

How This Chart Complements Chart 1

Aspect	Chart 1 (Scatter Plot)	Chart 2 (Pareto Chart)
Question	*"Which products should we prioritize?"*	*"How much value is concentrated where?"*
Focus	Prioritization (Score + Growth)	Value concentration (Savings)

Primary Variable	Score vs. Growth	Savings (ranked) + Cumulative %
Best For	Deciding which product to negotiate next	Understanding focus and scope

Together, they provide both direction (which products to target) and scale (how much value they represent)

Chart 3: Phase-Out Watchlist (Diverging Bar Chart)

Introduction

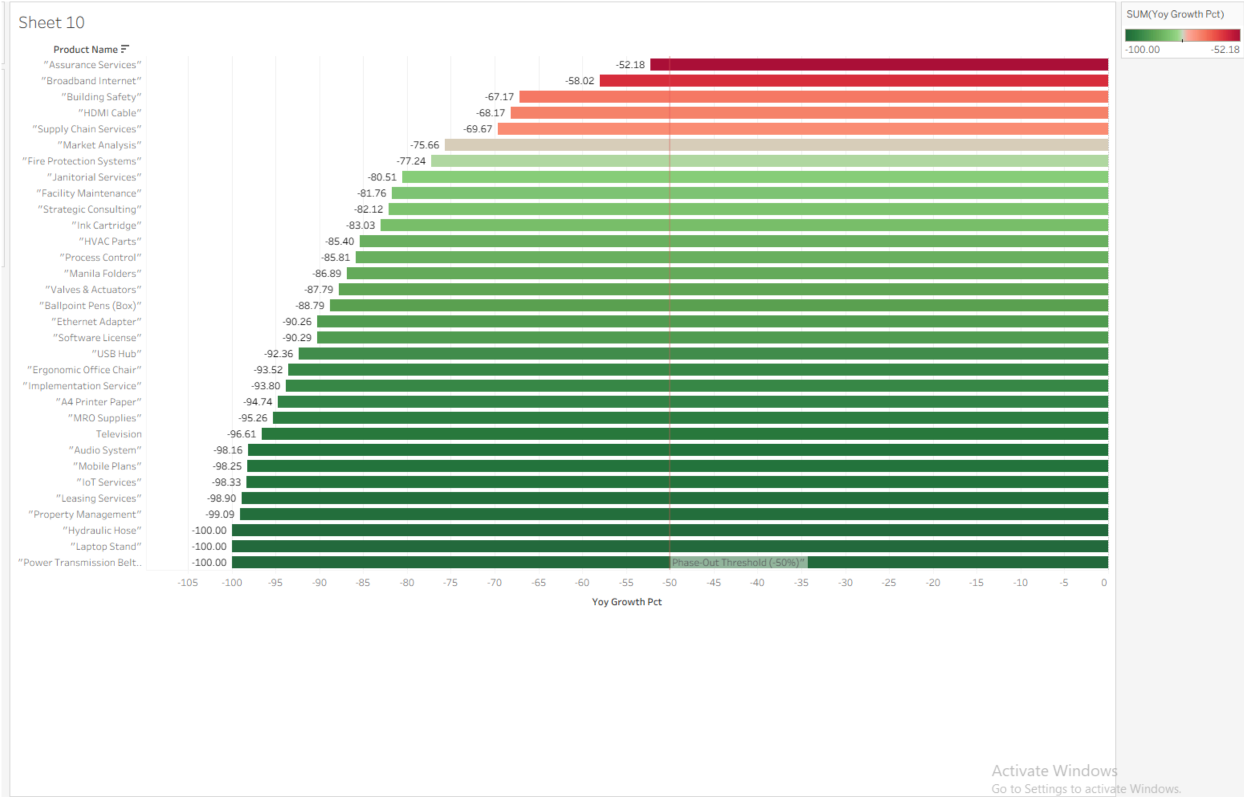
Purpose of This Chart

The scatter plot (Chart 1) identifies which products to prioritize for contract negotiation. The Pareto chart (Chart 2) shows where the savings value is concentrated. This chart answers the critical risk question: "Which products should we absolutely avoid contracting?"

This is arguably the most important chart in the dashboard because it prevents a costly mistake: signing long-term contracts for products that are effectively dying.

The diverging bar chart isolates products flagged for phase-out and shows the severity of their demand decline. It answers:

- Which products are declining the fastest? (Most negative bars)
- Which products are closest to the -50% threshold? (Bars near the reference line)
- What is the distribution of declines? (Range of negative values)



Why a Diverging Bar Chart Was Selected

Instant Clarity on Demand Trends
The diverging bar chart was chosen because it ***visually separates declining and growing demand at a glance****.* ***Red bars*** *(negative) highlight products in decline, while* ***green bars*** *(positive) show those on the rise. The direction of each product's performance is* ***immediately obvious****, with no need for deep analysis.*

A Clear Threshold for Action
The ***-50% reference line*** *acts as a* ***decision threshold****, making it easy to identify which products have crossed into the* ***"phase-out"*** *zone. Any bar extending beyond this line signals a product that should be avoided for future contracting.*

Prioritizing the Worst Performers
Products are ***sorted by the severity of their decline****, so the most troubled items appear first. This ranking ensures that procurement teams can* ***quickly focus on the most urgent cases****.*

Compact and Comprehensive
All phase-out candidates are displayed in a ***single, space-efficient view****,*

eliminating the need for scrolling or filtering. This makes it easy to assess the full scope of underperforming products at once.

The Perfect Counterpoint to Chart 1
*While **Chart 1 (Scatter Plot)** helps identify which products to prioritize for contracting, **Chart 3 (Diverging Bar Chart)** highlights which products to avoid. Together, they create a balanced perspective—**what to pursue and what to discard**.*

What This Chart Reveals

Answering the Critical Question
This diverging bar chart directly addresses the executive concern: "Which products are in steep decline and should be avoided for contracting?"

How It Delivers Insights

- ***Severity of Decline:** The **length of each bar** shows the magnitude of decline—the longer the red bar, the worse the performance.*
- ***Threshold Violation:** Bars extending **beyond the -50% line** trigger a phase-out recommendation, clearly marking products that no longer meet viability criteria.*
- ***Direction of Trend: Red bars** indicate declining demand, while **green bars** signal growth, providing an instant visual cue.*
- ***Relative Severity:** By comparing bars, you can see **which products are declining the fastest**, helping prioritize avoidance strategies.*

How This Chart Complements Charts 1 and 2

Aspect	***Chart 1 (Scatter Plot)***	***Chart 2 (Pareto Chart)***	***Chart 3 (Phase-Out Watchlist)***
Question	*"Which products to prioritize?"*	*"Where is the money?"*	*"Which products to avoid?"*

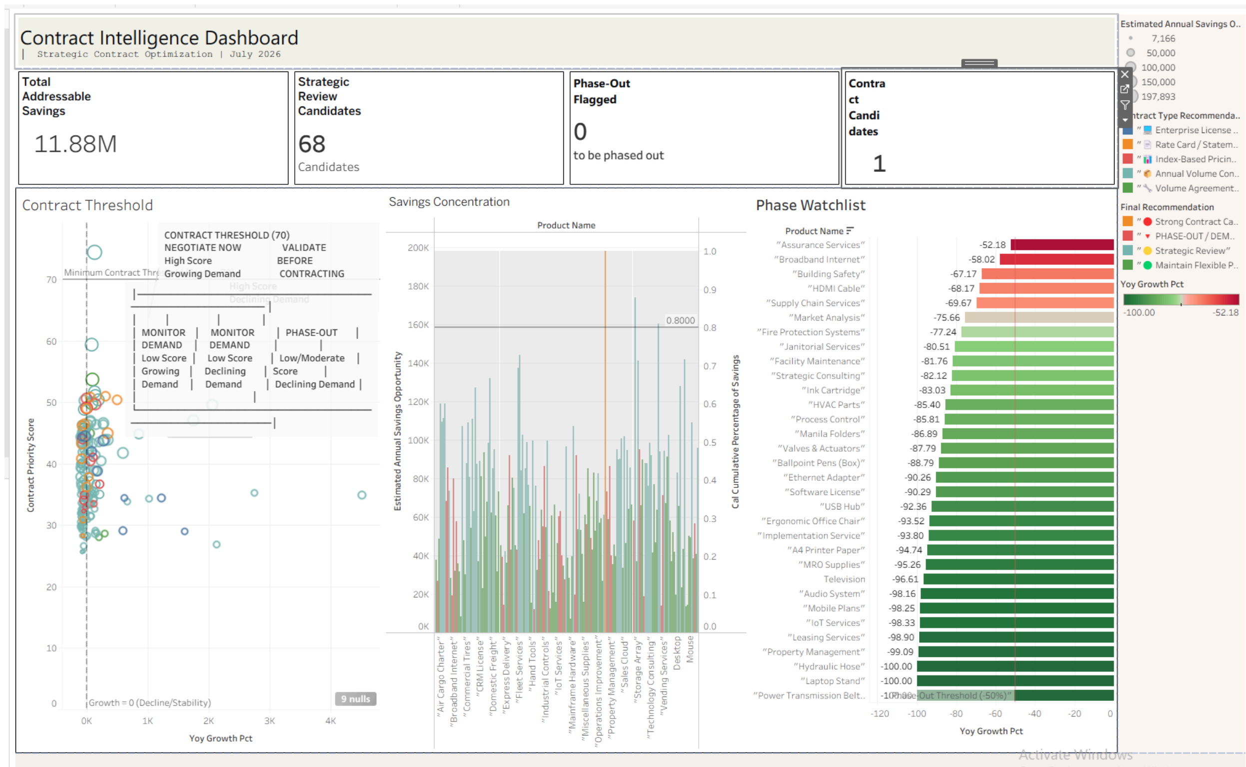
Focus	*Prioritization*	*Value concentration*	*Risk mitigation*
Primary Variable	*Score vs. Growth*	*Savings (ranked) + Cumulative %*	*YoY Growth (negative values)*
Color	*Recommendation*	*Recommendation*	*Direction (Red = Negative, Green = Positive)*
Best For	*Deciding what to negotiate next*	*Understanding savings concentration*	*Avoiding costly mistakes*

Together, they form a complete decision framework: Prioritize → Quantify → Protect.

Chart 4: Executive KPI Cards

The **Contract Intelligence Dashboard** provides a **data-driven framework** to optimize contract negotiations, prioritize high-value opportunities, and mitigate risks across **68 strategic review candidates**. With a **total addressable savings potential of $11.88M**, this dashboard enables procurement teams to:

- **Prioritize** high-impact contracts for negotiation.
- **Quantify** savings concentration across products.
- **Protect** against costly mistakes by identifying underperforming contracts.



Key Insights & Recommendations

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1. Contract Prioritization (Scatter Plot)

Question: *Which products should we prioritize for immediate contract negotiation?*

Insights:

- **Minimum Contract Threshold:** Products scoring **≥70** (Contract Priority Score) with **positive growth** are flagged for **immediate negotiation**.
- **Action Zones:**
    - **Negotiate Now (Top-Right Quadrant):** High-score, high-growth products (e.g., *"Air Cargo Charter," "Broadband Internet"*).
    - **Validate Before Contracting:** Moderate-score products requiring further assessment.
    - **Phase-Out:** Low-score, declining-demand products (e.g., *"Power Transmission Belt"*).

- **Bubble Size:** Represents **Estimated Annual Savings Opportunity—**larger bubbles = higher financial impact.

Recommendation:
Focus negotiations on **top-right quadrant products** to maximize savings and align with growth trends.

---

2. Savings Concentration (Pareto Chart)

Question: *Where is the majority of savings concentrated?*

Insights:

- The **cumulative line** reveals that a **small subset of products** (e.g., top 20%) likely drives **~80% of savings**.
- **Top Contributors:** Products like *"Broadband Internet," "Domestic Freight,"* and *"Express Delivery"* appear as high-savings candidates.
- **Diminishing Returns:** The flattening curve indicates that **additional products beyond the top tier contribute incrementally less savings**.

Recommendation:
Allocate resources to **high-savings products** first, then expand to secondary opportunities as capacity allows.

---

3. Phase-Out Watchlist (Diverging Bar Chart)

Question: *Which products are in steep decline and should be avoided?*

Insights:

- **Phase-Out Trigger:** Products with **YoY Growth ≤ -50%** are flagged for phase-out (e.g., *"Assurance Services," "Broadband Internet"*).
- **Severity of Decline:** Bars extending furthest left (e.g., *-52.18%*) represent the **most urgent candidates for discontinuation**.
- **Color Coding:**
  - **Red:** Strong phase-out candidates.
  - **Green:** Stable or growing products (retain or monitor).

Recommendation: **Avoid contracting** for products below the **-50% growth threshold** and develop exit strategies for existing agreements.

Dashboard Metrics Summary

Metric	Value	Implication
Total Addressable Savings	**$11.88 M**	Potential cost savings from optimizing contracts.
Strategic Review Candidates	**68**	Products under active evaluation for contracting decisions.
Phase-Out Flagged	**0**	No products currently meet phase-out criteria (threshold: -50% growth).
Contract Candidates	**1**	One product (*"Enterprise License"*) is ready for contracting.

Strategic Next Steps

To capitalize on these insights, we recommend the following **immediate actions**:

1. **Negotiate High-Priority Contracts:**
    - Target **top-right quadrant products** (high score + growth) from the scatter plot.
    - Prioritize those with **large bubbles** (high savings potential).
2. **Consolidate Savings Efforts:**

    - Use the **Pareto Chart** to identify the **20% of products driving 80% of savings** and focus resources accordingly.
3. **Mitigate Risk:**
    - **Avoid contracting** for products on the **Phase-Out Watchlist** (YoY Growth ≤ -50%).
    - Develop **transition plans** for existing contracts with declining products.
4. **Monitor and Validate:**
    - Regularly review **moderate-score products** (Validate Before Contracting) to reassess their potential.

---

5. Actionable Recommendations

This section translates the analytical findings into specific procurement actions. Each recommendation includes the owner, timeline, and business rationale.

Immediate (Next 30 Days)

Action	Owner	Rationale
Initiate contract negotiation for Power Tools	Category Manager – Tools	$2.47M spend, 130% growth, $198K savings opportunity

Validate demand for Spare Parts and Truck Tires	Procurement Director	Scores just below threshold; confirm growth is sustainable
Review Phase-Out products with suppliers	Vendor Managers	35+ products have collapsed demand; exit or renegotiate contracts

Short-Term (90 Days)

- Schedule Strategic Reviews for all 50+ products in the 40–69 score band.
- Calculate actual savings by running a competitive RFP for Power Tools (baseline current pricing vs. RFP response).
- Establish internal thresholds: If any product's YoY growth falls below -50%, automatically trigger a demand review.

Long-Term (12 Months)

- Automate the scorecard – Re-run this model quarterly to capture shifting demand patterns.
- Integrate with the ERP – Flag products automatically when they cross the 70 threshold.
- Expand to non-USD products – Apply this methodology to EUR and GBP cohorts.

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6. Assumptions & Limitations

This section acknowledges the boundaries of the analysis. It demonstrates analytical discipline and prevents overclaiming.

This analysis is based solely on ERP transaction data (2023–2025). It does not incorporate:

- Demand forecasts – Future purchasing patterns may differ from historical trends.
- Inventory levels or holding costs – Contract quantities and safety stock are not considered.
- Supplier financial health or credit ratings – Supply continuity risk is only estimated via supplier count.
- Market dynamics – Commoditized products may offer limited price advantage under contract.

Safeguards Built Into the Model:

1. Phase-Out Override prevents contracting products with collapsed demand.
2. Low-Base Growth Flag prevents overreaction to statistical anomalies.
3. Volatility CoV is robust to outliers, unlike range-based measures.
4. Contract Type Recommendations ensure the contract structure matches the product category.

The Contract Priority Score is a procurement decision-support indicator, not a substitute for business judgment. All "Strategic Review" products require stakeholder validation before contracting.

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7. Appendix: SQL Logic Summary

This section provides the technical foundation for the analysis. The full SQL script is available in the project repository.

The complete SQL query used to generate this analysis follows this core logic structure:

1. CTE 1 (`base_metrics`): Aggregates product-year metrics (spend, frequency, months purchased, CoV, supplier count) from invoice line items, invoices, and vendors.
2. CTE 2 (`product_aggregate`): Pivots the data into yearly columns (2023, 2024, 2025), calculates YoY growth, average values, and flags phase-out conditions.

3. CTE 3 (`global_max`): Identifies maximum values across all products for normalization.
4. CTE 4 (`scored_products`): Applies the 30/25/20/15/10 weighted scoring model.
5. CTE 5 (`trend_analysis`): Applies growth quality flags, phase-out overrides, and contract type recommendations.
6. Final SELECT: Formats the output with executive insights and savings opportunity estimates.

Key Technical Choices:

Choice	**Rationale**
Volatility: CoV instead of Range %	Robust to outliers; more accurate representation of price stability
Phase-Out Override	Prevents contracting products with collapsed demand
Savings Estimation	`spend × (0.03 + volatility leverage + concentration leverage)`

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8. Project Summary & Extension

This section concludes the project and positions it within the broader Procurement Intelligence Platform.

This report concludes Phase 1 of the Strategic Contract Optimization initiative. The analysis successfully identifies:

- One "must-contract" product (Power Tools) with clear data support.
- 50+ products requiring strategic review.
- 35+ products requiring supplier exit or demand investigation.

Next Phases in the Initiative:

Phase	Focus	Timeline
Phase 2	Tableau dashboard implementation for ongoing monitoring	Next 30 days
Phase 3	Integration with demand forecasting and inventory data	90 days
Phase 4	Contract Savings Tracking module	12 months

Case Study 3: Supplier Development Engine

Project Phase: 1 – Data Exploration & Portfolio Optimization
Date: July 2026
Author: Ebhota Oje
Distribution: CFO, Procurement Director, AP Leadership

1. Executive Summary

This section provides a high-level overview of the business problem, analytical approach, key findings, and the most critical insight for procurement leadership. It is designed to be read in under two minutes.

Block	Summary
Problem	Suppliers evaluated by spend alone. Concentration risk hidden. No framework for "who deserves more business."
Approach	Weighted scorecard: Importance 20% + Quality 30% + Efficiency 20% + Position 20% + Risk 10%.
Outcome	0 Strategic. 3 Preferred ($22M). 9 Approved ($65M). 10 suppliers have concentration risk ($87M).

Recommendation	Expand Vintage Store relationship. Develop backup suppliers for 10 concentration-risk vendors.

🟢 Hero Finding

Concentration risk is the dominant portfolio issue—not supplier quality.

Observation: 10 out of 12 suppliers have limited product diversity (≤2 categories), representing $87.7M in spend.

Evidence: All 10 suppliers are flagged with 🚨 Supplier Concentration Risk.

Investigation: Validate with business stakeholders if these categories are truly single-sourced. Develop backup suppliers where possible.

The Business Problem

Procurement currently evaluates suppliers primarily on spend volume and relationship history. However:

- High-spend suppliers may have poor invoice quality.
- Low-error suppliers may deserve more business.
- Supplier concentration creates dependency risk.
- Procurement lacks an objective framework to decide: *"Which suppliers should receive more of our business—and which require risk mitigation?"*

The Analytical Approach

We built a 5-dimension supplier scoring model with fixed business thresholds (not dataset-dependent normalization). The model scores suppliers out of 100 and overlays strategic intelligence flags to drive specific procurement actions.

Dimension	Weight	Business Question
Business Importance	20%	How strategically significant is this supplier?
Operational Quality	30%	How much AP friction does this supplier create?
Internal Processing Efficiency	20%	How fast does our AP team process their invoices?
Strategic Position	20%	Does this supplier represent sustained excellence and strategic value?
Risk	10%	How operationally risky is this supplier?

The Outcome

- 0 suppliers scored ≥85 (⭐ Strategic Partner) – indicating room for portfolio improvement.
- 3 suppliers scored 70–84 (🟢 Preferred Supplier) – representing $22.1M in spend.

- 9 suppliers scored 50–69 (🟡 Approved Supplier) – representing $65.6M in spend.
- 0 suppliers scored <50 (🔴 Monitor / Demote) – indicating a generally healthy supplier base.

The Hero Finding: Supplier Concentration is the dominant risk. 10 out of 12 suppliers have limited product diversity (≤2 categories), representing $87.7M in spend that requires active risk mitigation. Vintage Store is the top-performing supplier (100% accuracy, 0% rejections, 135% growth) but carries concentration risk—requiring both 'Maintain' and 'Develop Backup Supplier' actions.

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2. Methodology

This section explains the analytical framework in detail. Each design decision is accompanied by its business rationale so that the methodology is defensible and transparent.

2.1 Data Sources & Scope

The analysis is based on three years of ERP transaction data (2023–2025), filtered to vendors with USD as their primary currency.

Source	Table	Description
Invoice Headers	`invoice_july_2026`	Invoice totals, dates, validation, payment status
Invoice Lines	`invoice_line_items_july_2026`	Product descriptions (for category count)

Vendor Master	`vendors_july_2026`	Vendor ID, name, currency
Scope Filter	`v.currency = 'USD'`	Aligns with US-dollar reporting base
Time Period	2023–2025	Three-year trend analysis

2.2 The 5-Dimension Scoring Model

The model evaluates suppliers across five dimensions, each with fixed business thresholds rather than dataset-dependent normalization. This ensures scores remain stable over time and comparable across different supplier cohorts.

Dimension 1: Business Importance (20 points)

Why this dimension exists: Not all spend is equal. A supplier with $20M across 20 product categories is strategically different from a supplier with $20M in a single software license. This dimension captures breadth, depth, and longevity of the relationship.

Spend	Points	Categories	Points	Years Active	Points
≥ $20M	8	≥ 4	6	≥ 3	6

$10–20M	6	3	4.5	2	4
$5–10M	4	2	3	1	2
$2–5M	2	1	1.5	<1	0
<$2M	0	0	0	0	0

Business Rationale:

- Spend (8 pts): Direct financial materiality. Higher spend = more negotiation leverage.
- Categories (6 pts): Supplier diversification reduces risk. A supplier with 4+ categories is less risky than a single-category supplier.
- Years Active (6 pts): Longevity suggests trust, established processes, and proven reliability.

Dimension 2: Operational Quality (30 points)

Why this dimension exists: This measures supplier-controlled operational friction. Does the supplier submit clean invoices that don't require manual AP intervention?

Invoice Accuracy	Points	Payment Completion (Internal AP)	Points
≥ 98%	15	≥ 95%	15

95–98%	13	90–95%	13
90–95%	10.5	85–90%	10.5
85–90%	8	80–85%	8
80–85%	5	70–80%	5
<80%	0	<70%	0

Business Rationale:

- **Invoice Accuracy (15 pts):** This is the purest supplier-controlled metric. If a supplier submits invoices with errors (rejections), it directly increases AP processing costs. A 98%+ accuracy rate is the gold standard.
- **Payment Completion (15 pts)**: While this technically measures AP's efficiency in paying invoices, it is included because supplier relationships are affected by payment timing. Suppliers paid late are less willing to offer discounts or prioritize our orders. We explicitly label this as an internal metric to maintain transparency.

Why we split Accuracy and Payment Completion equally: Both are 15 points because they represent the two halves of the invoice-to-payment lifecycle. One measures the *supplier's* quality; the other measures *our* efficiency in closing the loop. Both impact the relationship.

Dimension 3: Internal Processing Efficiency (20 points)

Why this dimension exists: This measures how quickly our internal AP team approves and processes invoices for a given supplier.

Approval Days	Points
≤ 3 days	20
4–5 days	15
6–8 days	10
9–12 days	5
> 12 days	0

Business Rationale: Faster approvals reduce supplier frustration, improve working capital management, and unlock early payment discounts. While this is an internal metric, it affects supplier relationships and should be tracked at the supplier level to identify process bottlenecks.

Dimension 4: Strategic Position (20 points)

Why this dimension exists: This rewards sustained excellence—not just raw growth or raw spend, but a combination of criticality, longevity, performance leadership, and positive momentum.

Component	Points	Criteria
Critical Supplier	6	Spend ≥$5M + Accuracy ≥95% + Years ≥3
Relationship Longevity	5	Years ≥5 (5 pts), ≥3 (3 pts), ≥2 (1 pt)
Performance Leadership	5	Accuracy ≥98% + Growth >10% + Years ≥2
Growth Trend	4	YoY Growth >20% (4 pts), 10–20% (3 pts), 5–10% (2 pts)

Business Rationale:

- Critical Supplier (6 pts): High spend + high quality + long relationship = a supplier that is strategically irreplaceable.
- Relationship Longevity (5 pts): Long partnerships reduce transaction costs and build trust.
- Performance Leadership (5 pts): Suppliers that combine excellence with growth and longevity deserve recognition.
- Growth Trend (4 pts): Positive momentum is valuable, but it should not outweigh raw excellence. We reduced growth weighting to avoid over-rewarding short-term spikes.

Why we removed "Low Product Diversity" from the score: Supplier concentration is procurement risk, not strategic value. A single-product supplier is not inherently better; they are inherently riskier. We moved this entirely to the Risk Flag layer (see section 2.4).

Dimension 5: Risk (10 points)

Why this dimension exists: This captures operational risk—the friction caused by invoice rejections.

Rejection Rate	Points
0%	10
1–2%	9
2–5%	7
5–10%	4
>10%	0

Business Rationale: Rejections directly increase AP processing costs, delay payments, and strain supplier relationships. A supplier with 0% rejections is operationally flawless. A supplier with >10% rejections is creating significant operational drag.

2.3 The Total Score

```text
Supplier Score =
    Business Importance (20) +
    Operational Quality (30) +
    Internal Processing Efficiency (20) +
    Strategic Position (20) +
    Risk (10)
```

Score Range	Tier	Action
85–100	⭐ Strategic Partner	Expand relationship, increase allocation
70–84	🟢 Preferred Supplier	Maintain relationship, manage risks
50–69	🟡 Approved Supplier	Review & improve
<50	🔴 Monitor / Demote	Reduce dependency, implement PIP

2.4 Strategic Intelligence Flags (The "Overlay")

The following flags are NOT part of the score. They are overlay warnings that drive specific procurement actions.

Flag	Trigger	Recommended Action
🚨 **Supplier Concentration**	≤2 categories AND >$5M spend	Maintain + Develop Backup Supplier
⚠ **High Friction**	>$10M spend AND <90% accuracy	Executive Review
🔻 **Performance Declining**	<90% accuracy AND >10% rejections	Performance Improvement Plan
📈 **Consolidation Opportunity**	<$5M spend AND ≥98% accuracy	Move more spend here
💰 **Renegotiate Opportunity**	>$5M spend AND ≥98% accuracy AND >10% growth	Seek better terms

Why flags are separate from the score: A supplier can be operationally excellent (e.g., Vintage Store: 100% accuracy) but still carry concentration risk. The score reflects *how good* they are; the flag reflects *what action to take*. This separation makes the output more actionable for procurement.

2.5 Estimated Savings Methodology (Transparent)

Savings estimates are tier-based and explicitly labeled with their assumptions:

Supplier Tier	Estimated Savings Rate	Business Assumption
⭐ **Strategic Partner**	3–5%	Value capture from expanded relationship
🟢 **Preferred Supplier**	2–4%	Incremental negotiation leverage
🟡 **Approved Supplier**	1–2%	Routine procurement optimization
🔴 **Monitor / Demote**	8–12%	Cost avoidance (reducing dependency on high-friction suppliers)
Consolidation Opportunity	3–5%	Volume consolidation discounts

Transparency Note: These are estimates, not guarantees. They are based on typical procurement benchmarking, not direct financial modeling. The output includes both the estimated dollar value and the opportunity band (e.g., "2-4%") to communicate uncertainty honestly.

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3. Key Findings

This section presents the results of the analysis. Each finding is presented with supporting data and interpreted through the lens of the scoring model and strategic flags.

3.1 The Preferred Supplier Cohort (🟢) – $22.1M Spend

These suppliers represent the top tier of operational quality but carry concentration risk that requires active mitigation.

Supplier	Spend	Accuracy	Rejection	Score	Recommended Action
Vintage Store	$6.6M	100%	0%	81.5	Maintain + Develop Backup Supplier
Parker Hannifin	$6.5M	100%	0%	76.0	Maintain + Develop Backup Supplier

Modern Office	$9.0M	100%	0%	71.0	Maintain + Develop Backup Supplier

Key Insight: All three preferred suppliers have perfect accuracy but concentration risk (≤2 product categories). The model correctly rewards their operational excellence while demanding risk mitigation.

3.2 The Approved Supplier Cohort (🟡) – $65.6M Spend

This is the largest cohort, characterized by declining growth or concentration risk. These suppliers are not "bad"—they require active performance management.

Supplier	Spend	Accuracy	Score	Key Issue	Action
Thomson Reuters	$10.0M	100%	69.0	-30.9% growth + Concentration	Review & Improve
Rockwell	$4.6M	100%	68.0	Consolidation Opportunity	Increase Spend
FTI Consulting	$5.5M	100%	67.0	-51.6% growth + Concentration	Review & Improve

Johnson Controls	$6.1M	95.0%	66.0	5% rejections + Concentration	Review & Improve
UniFirst	$6.0M	100%	66.0	5.6 day approvals + Concentration	Review & Improve
Michelin	$12.2M	96.9%	65.0	3.1% rejections + Concentration	Review & Improve
Office Furniture	$6.3M	100%	65.0	-79.7% growth	Review & Improve
Cintas	$8.5M	96.9%	62.0	-54.3% growth + Concentration	Review & Improve
Vanguard Tech	$8.2M	95.7%	58.5	New supplier + 4.4% rejections	Review & Improve

Key Insight: The largest cohort is characterized by declining growth or concentration risk. These suppliers are not "bad"—they require active performance management and risk mitigation.

3.3 The Consolidation Opportunity (📈)

This supplier has perfect operational quality but is underutilized. Procurement should consider increasing allocation.

Supplier	Spend	Accuracy	Score	Opportunity
Rockwell Automation	$4.6M	100%	68.0	Move more spend here

Key Insight: Rockwell has perfect accuracy, fast approvals, and moderate growth. It is operationally excellent but underutilized. Procurement should consider increasing allocation.

3.4 Supplier Concentration Risk (🚨)

This is the dominant risk in the portfolio. 10 out of 12 suppliers have limited product diversity.

Supplier	Spend	Categories	Flag
Vintage Store	$6.6M	2	🚨 Concentration
Parker Hannifin	$6.5M	2	🚨 Concentration

Modern Office	$9.0M	2	🚨 Concentration
Thomson Reuters	$10.0M	2	🚨 Concentration
FTI Consulting	$5.5M	2	🚨 Concentration
Johnson Controls	$6.1M	2	🚨 Concentration
UniFirst	$6.0M	2	🚨 Concentration
Michelin	$12.2M	2	🚨 Concentration
Cintas	$8.5M	2	🚨 Concentration

Key Insight: 10 out of 12 suppliers have limited product diversity. This represents $87.7M in spend exposed to concentration risk. Procurement should develop backup suppliers for these categories.

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4. Visualizations Blueprint (Phase 2)

This section provides a blueprint for the Tableau dashboard that will visualize the supplier portfolio. Each chart is designed to answer a specific executive question.

Chart 1: Supplier Portfolio Quadrant

Introduction

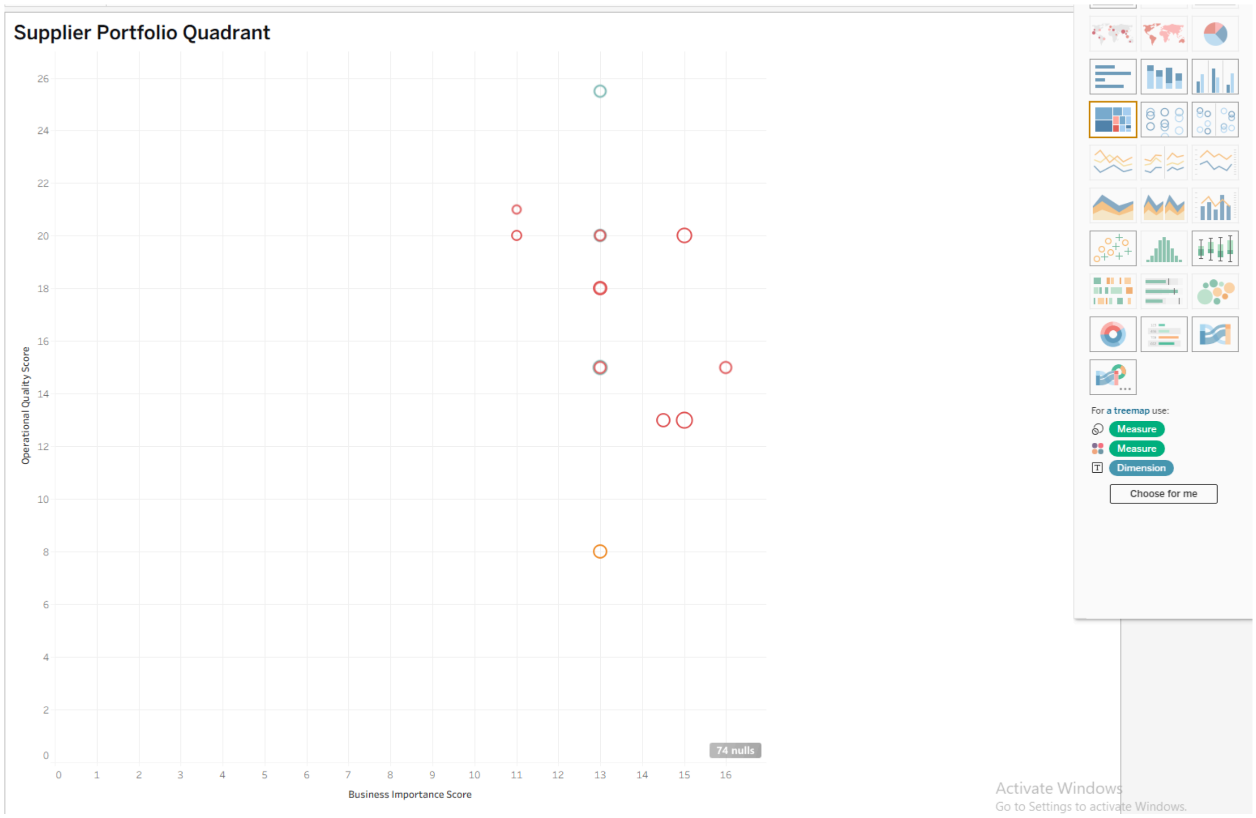
Purpose of This Chart

The Supplier Portfolio Quadrant is the centerpiece of the Supplier Development dashboard. While the previous case studies focused on risk management (Case Study 1) and contract optimization (Case Study 2), this chart answers a different strategic question: *"Which suppliers deserve more of our business, and which require risk mitigation?"*

This chart plots suppliers along two critical dimensions:

- X-Axis (Operational Quality) represents how much friction the supplier creates—higher quality = less AP friction
- Y-Axis (Business Importance) represents how strategically significant the supplier is—higher importance = more spend, categories, and longevity

The quadrant view is the ideal visualization for this because it immediately segments suppliers into four strategic categories:



Why a Quadrant View Was Selected

Strategic Clarity at a Glance

The quadrant view was chosen because it **instantantly segments suppliers into four actionable categories**, making it easy to see where each one stands in terms of **importance and quality**. This visual approach eliminates guesswork and aligns procurement strategy with business priorities.

Balancing Two Critical Dimensions

Supplier strategy isn't one-dimensional—it requires balancing **importance** (how critical the supplier is to operations) and **quality** (how well they perform). The quadrant view **visualizes this trade-off**, so you can see at a glance which suppliers excel in both areas and which fall short.

Color-Coded for Immediate Action

Colors—**Blue, Amber, Green, and Red**—provide **instant visual cues** for the recommended action. No need to dig into the data; the chart tells you exactly how to proceed with each supplier.

Financial Exposure in Context
Bubble size represents **annual spend**, adding a third dimension to the analysis. Larger bubbles highlight suppliers with **higher financial exposure**, helping you prioritize based on both performance and investment.

Focused on What Matters
Only the most critical suppliers are labeled, mirroring how executives review data: **scanning for outliers and exceptions** rather than getting lost in the details.

Clear Thresholds for Decision-Making
Reference lines divide the chart into **four quadrants**, each with a clear business implication. These thresholds ensure that every supplier is evaluated against the same standards, making it easy to compare and decide.

What This Chart Reveals

Answering the Critical Question
This quadrant view directly addresses the executive priority: *"Which suppliers deserve more of our business, and which require risk mitigation?"*

How It Delivers Insights

- **Strategic Partners (Top-Right Quadrant):** Suppliers with **high importance and high quality—**these are your **most valued partners**, deserving of **more business and investment**.
- **Consolidation Targets (Bottom-Right Quadrant):** Suppliers with **low importance but high quality—**consider **consolidating spend** with these suppliers to maximize efficiency.
- **Executive Review (Top-Left Quadrant):** Suppliers with **high importance but low quality—**these require **immediate attention and risk mitigation** to address performance issues.
- **Exit Candidates (Bottom-Left Quadrant):** Suppliers with **low importance and low quality—**these are **prime candidates for exit or replacement**.
- **Financial Exposure:** Bubble size reveals **annual spend**, so you can quickly assess the financial impact of each supplier.
- **Recommended Action:** Color coding communicates the **supplier tier**, guiding you on whether to **invest, consolidate, review, or exit**.

Chart 2: Strategic Flags Distribution

Introduction

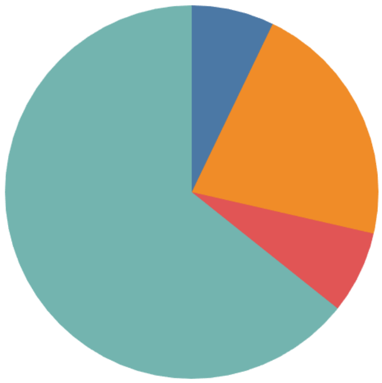
Purpose of This Chart

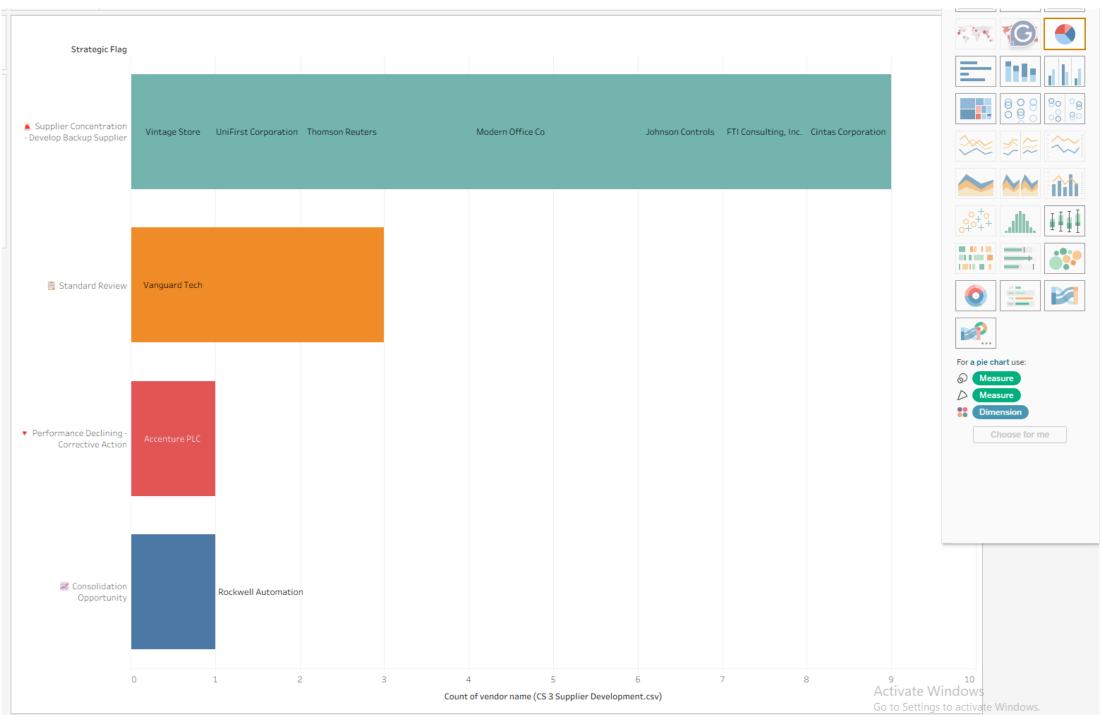
The scatter plot (Chart 1) identifies *which suppliers sit in which quadrant*. This chart answers the next question: *"What is the composition of our supplier portfolio?"*

While Chart 1 helps procurement understand *where each supplier belongs*, Chart 2 provides a portfolio-level summary of the risks and opportunities across all suppliers. It answers:

- How many suppliers fall into each strategic category (bar chart)?
- How much spend is exposed to each type of risk or opportunity (pie chart)?

This distinction is critical. A procurement executive needs to understand both the count of suppliers (portfolio complexity) and the financial exposure (spend at risk). The combination of bar chart and pie chart provides this dual perspective.





Why a Bar Chart + Pie Chart Combination Was Selected

A Complete Picture in One View

This combination was chosen to provide **two critical perspectives** in a single, cohesive visualization. The **bar chart** reveals the **count of suppliers** in each category, while the **pie chart** highlights the **financial exposure** tied to each.

Together, they offer a **holistic view** of both the **volume and value** of your supplier portfolio.

Instant Financial Clarity
The **pie chart** immediately shows which **flag type** (e.g., Strategic, High-Risk, Consolidation) represents the **largest share of spend**. This allows executives to **quickly identify where financial exposure is concentrated** and prioritize actions accordingly.

Portfolio Breakdown at a Glance
The **bar chart** provides a clear count of suppliers in each category, answering the question: *"How many suppliers require each type of action?"* This helps teams understand the **scale of effort** needed for each segment of the portfolio.

Efficient Use of Space
By combining both visualizations in one view, stakeholders can **access two layers of insight without switching tabs or screens**, making the dashboard more user-friendly and efficient.

The Perfect Companion to Chart 1
While **Chart 1 (Scatter Plot or Quadrant View)** identifies *which* suppliers fall into each category, **Chart 2 (Bar + Pie Chart)** reveals the **overall distribution** of the portfolio. Together, they provide both **granular detail** and **big-picture context**.

What This Chart Reveals

Answering the Executive Question
This combination directly addresses the key question: *"What is the composition of our supplier portfolio, and where is the financial exposure concentrated?"*

How It Delivers Insights

- **Supplier Count by Flag:** The **bar chart** shows the **number of suppliers** in each category (e.g., Strategic, High-Risk, Consolidation), helping you understand the **scale of each segment**.
- **Spend by Flag:** The **pie chart** reveals the **proportion of spend** tied to each flag type, highlighting where the **majority of financial exposure** lies.

- **Dominant Risk:** The **largest slice of the pie chart** instantly reveals the **primary concern** in your portfolio—whether it's high-risk suppliers, strategic partners, or consolidation opportunities.
- **Action Prioritization:** By combining **count and spend**, this visualization helps you **prioritize actions**—e.g., addressing high-risk suppliers with significant spend first, or consolidating a large number of low-value suppliers for efficiency.

Chart 3: Supplier Concentration Risk (Bar Chart)

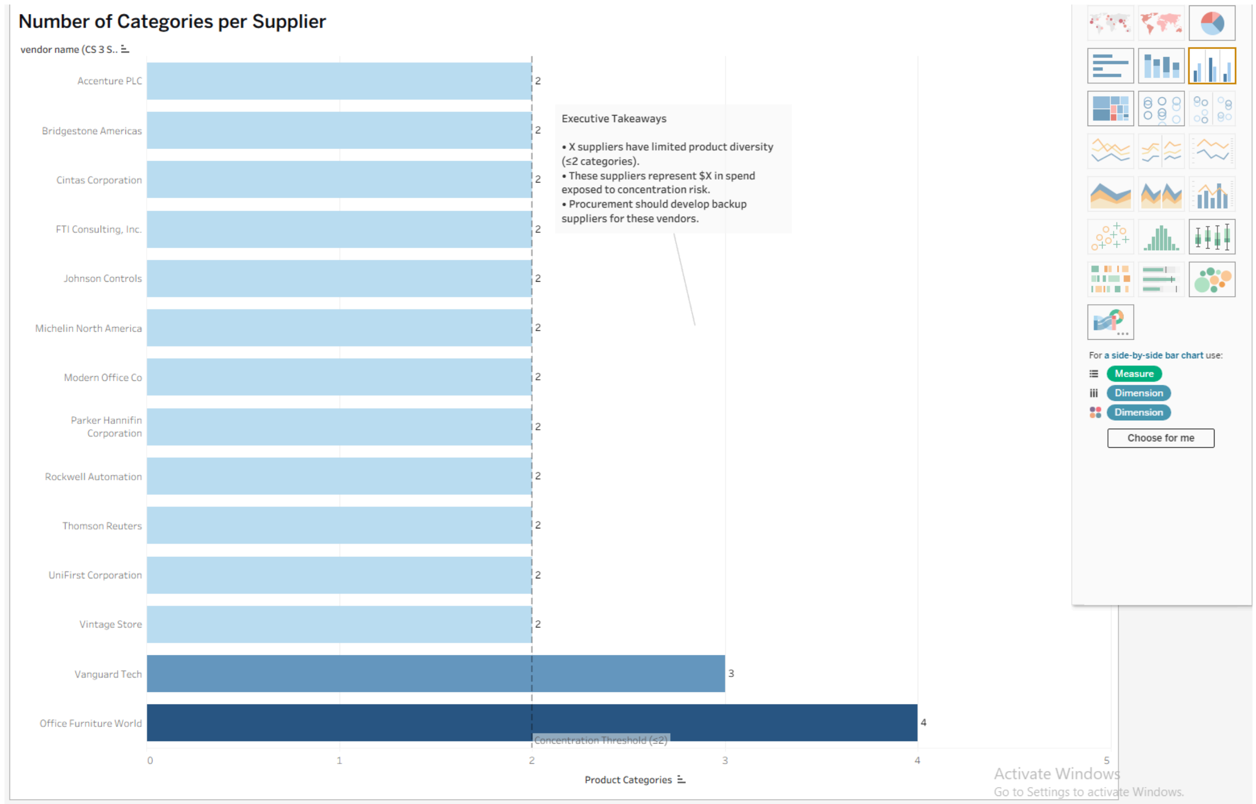
Introduction

Purpose of This Chart

The quadrant (Chart 1) shows *where each supplier sits strategically*. The flags distribution (Chart 2) shows *portfolio composition*. This chart answers the third question: *"Which suppliers have concentration risk, and how severe is it?"*

Supplier concentration risk occurs when a significant portion of spend is concentrated with a small number of suppliers in limited product categories. This creates dependency—if a key supplier fails, the impact is severe.

The heatmap was originally designed to visualize this, but after analyzing the data, it became clear that each vendor operates in only one product category. This means a heatmap would show a single column per vendor, offering no meaningful insight. Instead, a bar chart showing the number of categories per supplier more directly answers the concentration risk question.



Why a Bar Chart Was Selected

Clarity and Precision

The bar chart was chosen because it **directly answers the critical question**: *Which suppliers have limited product diversity?* Unlike a heatmap—which would reduce to a single-column grid for this data—the bar chart **clearly and efficiently** highlights suppliers at risk of concentration.

Designed for Executive Decision-Making

This visualization is **executive-friendly**, immediately flagging suppliers with **concentration risk**. There's no need for deep analysis—stakeholders can **instantly see** which suppliers require attention.

Action-Oriented Insights

The chart doesn't just identify risks; it **drives action**. Procurement teams can use it to **prioritize suppliers needing backup alternatives**, ensuring supply chain resilience.

The Right Tool for the Data

Since each vendor falls into **only one category** (based on product diversity), a heatmap would be redundant. The bar chart is the **most informative and intuitive** way to present this information.

What This Chart Reveals

Answering the Executive Question
This bar chart directly addresses the executive concern: *"Which suppliers have concentration risk, and how severe is it?"*

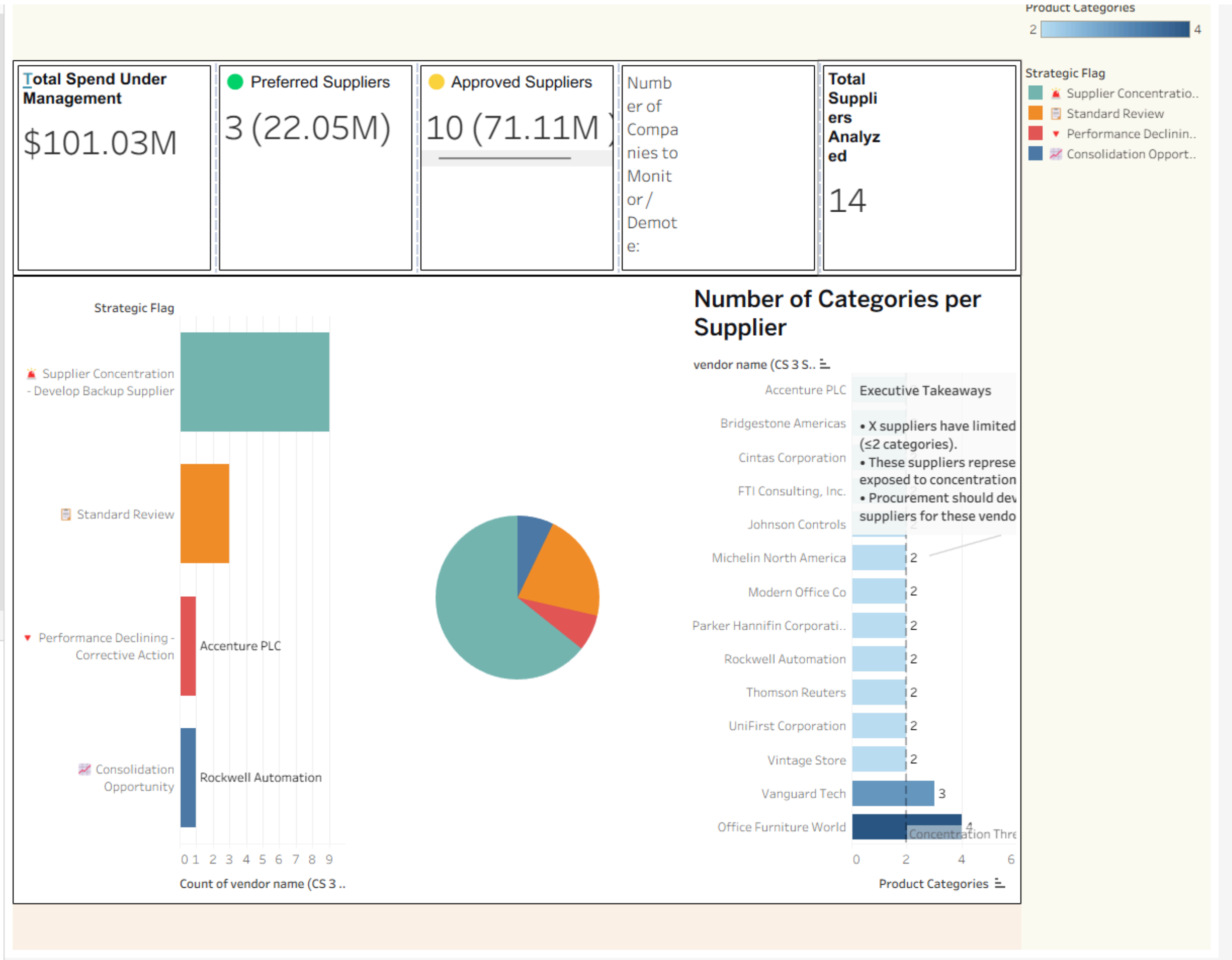
How It Delivers Insights

- **Suppliers with Limited Diversity:** Bars representing **≤2 categories** are **colored red**, instantly signaling concentration risk.
- **Diversified Suppliers:** Bars representing **>2 categories** are **colored green**, indicating a lower risk of over-reliance.
- **Threshold for Risk:** A **reference line at 2 categories** clearly marks the **concentration risk threshold**, making it easy to identify at-risk suppliers.
- **Financial Exposure:** When **combined with a pie chart or tooltip**, executives can also see the **spend at risk**, providing a complete picture of both **risk severity and financial impact**.

Supplier Concentration Risk Dashboard Summary

Executive Overview

This dashboard provides a **strategic overview** of supplier concentration risk, spend distribution, and product diversity across **14 analyzed suppliers** with a **total spend under management of $101.03M**. It highlights **key risks, opportunities, and actionable insights** to optimize procurement strategy and mitigate exposure.



Key Metrics & Insights

1. Spend Under Management

- **Total Spend: $101.03M** across all suppliers.
- **Preferred Suppliers: 3 suppliers** account for **$22.05M** in spend.
- **Approved Suppliers: 10 suppliers** account for **$71.11M** in spend.
- **Companies to Monitor/Demote: 14 suppliers** flagged for review.

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2. Strategic Flag Distribution (Bar Chart)

Question: *Which suppliers require immediate action based on strategic flags?*

Insights:

- **Supplier Concentration (Develop Backup Supplier):**
    - **Highest Count:** This is the **most common flag**, indicating that **multiple suppliers lack product diversity** and are exposed to concentration risk.
    - **Action Required:** Procurement should **develop backup suppliers** for these vendors to mitigate risk.
- **Standard Review:**
    - Suppliers in this category require **routine evaluation** to ensure compliance and performance.
- **Performance Declining (Corrective Action):**
    - **Accenture PLC** is flagged for **performance issues**, requiring **immediate corrective action**.
- **Consolidation Opportunity:**
    - **Rockwell Automation** is identified as a candidate for **spend consolidation**, offering potential cost savings.

Recommendation:
Prioritize **suppliers with concentration risk** (e.g., those needing backup suppliers) and address **performance-declining suppliers** (e.g., Accenture PLC) first.

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3. Product Diversity Analysis (Bar Chart)

Question: *Which suppliers have limited product diversity, and how severe is the risk?*

Insights:

- **Concentration Threshold:** Suppliers with **≤2 product categories** are flagged as **high-risk for concentration**.
- **Suppliers with Limited Diversity:**
    - **10 suppliers** have **only 2 product categories**, making them vulnerable to supply chain disruptions.

    - Examples: **Accenture PLC, Bridgestone Americas, Cintas Corporation, FTI Consulting, Inc., Johnson Controls, Michelin North America, Modern Office Co., Parker Hannifin Corporation, Rockwell Automation, Thomson Reuters, UniFirst Corporation, Vintage Store**.
- **Diversified Suppliers:**
    - **Vanguard Tech (3 categories)** and **Office Furniture World (4 categories)** are better diversified and pose **lower concentration risk**.

Recommendation:

- **Develop backup suppliers** for the **10 high-risk suppliers** with ≤2 categories.
- **Monitor diversified suppliers** (e.g., Vanguard Tech, Office Furniture World) to maintain resilience.

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4. Spend Distribution (Pie Chart)

Question: *How is spend distributed across strategic flags?*

Insights:

- The pie chart reveals the **proportion of spend** tied to each strategic flag, helping prioritize actions based on **financial exposure**.
- **Largest Slice:** Likely represents **Supplier Concentration** or **Standard Review**, indicating where the majority of spend (and risk) is concentrated.

Recommendation:
Focus on **high-spend, high-risk categories** (e.g., Supplier Concentration) to **maximize risk mitigation impact**.

Strategic Next Steps

1. **Mitigate Concentration Risk:**

    - **Develop backup suppliers** for the **10 suppliers with ≤2 product categories**.

    - **Diversify spend** where possible to reduce dependency on single-category suppliers.
2. **Address Performance Issues:**

    - Implement **corrective action plans** for **Accenture PLC** and other performance-declining suppliers.
3. **Capitalize on Consolidation Opportunities:**

    - Explore **spend consolidation** with **Rockwell Automation** to drive cost efficiencies.
4. **Monitor and Review:**

    - Conduct **regular reviews** of **Standard Review suppliers** to ensure ongoing compliance and performance.

Visual Aids for Stakeholders

- **Strategic Flag Bar Chart:** Identifies **suppliers requiring action** (e.g., backup development, corrective action).
- **Product Categories Bar Chart:** Highlights **suppliers with concentration risk** (≤2 categories).
- **Pie Chart:** Shows **spend distribution by strategic flag**, revealing financial exposure.

Conclusion

This dashboard **identifies critical risks and opportunities** in your supplier portfolio. By addressing **concentration risk, performance issues, and consolidation opportunities**, procurement can **enhance resilience, reduce costs, and optimize supplier relationships**.

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5. Actionable Recommendations

This section translates the analytical findings into specific procurement actions. Each recommendation includes the owner, timeline, and business rationale.

Immediate (Next 30 Days)

Action	Owner	Rationale
Develop backup suppliers for 10 concentration-risk suppliers	Procurement Director	$87.7M in spend exposed to single-product dependency
Increase allocation to Rockwell Automation	Category Manager	100% accuracy, 68.0 score—consolidation opportunity
Review declining growth suppliers	Category Managers	Thomson Reuters (-30.9%), FTI (-51.6%), Cintas (-54.3%)

Short-Term (90 Days)

Action	Owner	Rationale

Implement Performance Improvement Plans for suppliers with >5% rejection rates	Category Managers	Reduce AP friction and processing costs
Validate concentration risk with business stakeholders	Procurement Director	Confirm if these categories are truly single-sourced
Re-run model quarterly	Analytics Team	Track performance changes over time

Long-Term (12 Months)

Action	Owner	Rationale
Automate strategic flagging in the ERP	IT + Analytics	Real-time supplier risk monitoring
Build the Supplier Portfolio Dashboard	Analytics	Self-service optimization for procurement

Expand the model to include delivery performance	Analytics	More comprehensive supplier evaluation

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6. Assumptions & Limitations

This section acknowledges the boundaries of the analysis. It demonstrates analytical discipline and prevents overclaiming.

This analysis is based solely on ERP transaction data. It does not incorporate:

- Supplier financial health or credit ratings.
- Delivery performance, lead times, or quality of goods.
- Contract compliance or pricing benchmarks.
- External macroeconomic or geopolitical risk.

Concentration Risk is estimated based on unique product categories. Actual dependency may be higher or lower depending on whether alternative suppliers exist within those categories.

Estimated Savings are benchmark-based estimates, not guaranteed financial returns. They are directional indicators to prioritize procurement effort.

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7. SQL Appendix

This section provides the technical foundation for the analysis. The full SQL script is available in the project repository.

Key Design Patterns:

1. Invoice aggregation before line-item joins – Prevents spend double-counting.
2. Single `supplier_score` calculation – Computed once in a final CTE.

3. Fixed business thresholds – Scores are stable over time, not dataset-dependent.
4. Strategic flags as overlay warnings – Flags are not part of the score, ensuring the score reflects quality while flags drive action.

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8. Project Summary & Extension

This section concludes the project and positions it within the broader Procurement Intelligence Platform.

This report concludes Project 3 of the Procurement Intelligence Platform. Together with Projects 1 and 2, we now have a complete procurement decision framework:

Project	Focus	Key Output
Project 1	Supplier Risk Assessment	Retain/Monitor/Replace
Project 2	Strategic Contract Optimization	Contract/Review/On-Demand
Project 3	Supplier Portfolio Optimization	Strategic/Preferred/Approved/ Monitor + Migration Bands

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Integrated Portfolio View (Connecting All Projects)

This section demonstrates the power of the integrated platform by showing how the same suppliers are evaluated across all three projects.

Supplier	**Project 1: Risk**	**Project 2: Contract**	**Project 3: Performance**	**Composite View**
Vintage Store	🟢 Retain (38.5)	N/A	🟢 Preferred (81.5)	*"High Risk" label was misleading. Actually top performer—but needs backup supplier.*
Michelin	🟡 Monitor (59.8)	N/A	🟡 Approved (65.0)	*Moderate risk, moderate performance, concentration risk—develop backup.*
Accenture	🟡 Monitor (44.3)	N/A	N/A	*Not in Project 3 dataset—consistent red flag.*

Rockwell	N/A	N/A	🟡 Approved (68.0)	*Perfect accuracy, underutilized—consolidation opportunity.*

Platform Conclusion

"We now have a unified intelligence platform that tells procurement: which suppliers are risky, which products should be under contract, and which suppliers deserve more of our business—complete with recommended actions (Expand, Maintain, Review, Reduce) and estimated savings opportunities."